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Full-Stock-Payment Marginalization in Merger and Acquisition Transactions

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Abstract. The number of merger and acquisition (M&A) transactions paid fully in stock in the U.S. market declined sharply after 2001, when pooling and goodwill amortization were abolished by the Financial Accounting Standards Board. Did this accounting rule change really have such far reaching implications? Using a difference-in-differences test and Canada as a counterfactual, this study reveals that it did. We also report several other results confirming the role of pooling abolishment, including (i) that the decrease in full stock payment relates to CEO incentives and (ii) that previously documented determinants of the M&A mode of payment cannot explain the post-pooling abolishment pattern. These results are also robust to controls for various factors, such as the Internet bubble, the exclusion of cross-border deals, the presence of Canadian cross-listed firms, the use of a constant sample of acquirers across the pooling and post-pooling abolishment periods, the use of Europe as an alternative counterfactual, and controls for the SEC Rule 10b-18 share repurchase safe harbor amendments of 2003.

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Keywords: mergers and acquisitions • method of payment • pooling of interest • purchase method

1. Introduction

Over the past 25 years, U.S. merger and acquisition (M&A) transactions that have been fully paid in stock have displayed a striking pattern: from half of all transactions characterized by this payment mode during the 1990s, the percentage has fallen to around 10% in recent years (value-based percentages). This long-term evolution has mostly remained unnoticed in the academic community.¹ We investigate the causes of this long-term evolution.

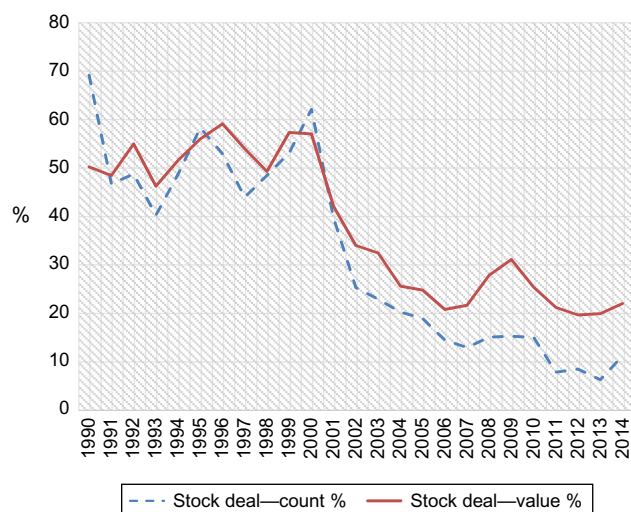
There has been much research into payment mode choices in M&A transactions in corporate finance because this topic offers a fertile ground for testing theories and developing a modern view of the firm. Betton et al. (2008) identify four factors that drive payment mode choices: taxes (Section 368 of the U.S. Internal Revenue Code), information asymmetries (Eckbo et al. 1990, Fishman 1989, Hansen 1987), capital structure and control (Harris and Raviv 1988, Stulz 1988), and agency-based and behavioral arguments (Jensen 2005, Rhodes-Kropf and Viswanathan 2004, Shleifer and Vishny 2003). Considerable empirical literature has tested the corresponding predictions. For example, in early research, Travlos (1987) established that equity transaction announcements generated negative market reactions of about 1% for acquisitions of public companies. More recently, Fuller et al. (2002) show that market reactions to stock-paid transactions depend on

the consideration offered but also on the target's status, such that they generate positive market reactions if the target is a private company.

The long-term marginalization of fully stock-paid M&A transactions offers a unique context for investigating whether classic payment mode theories represent first-order drivers of payment mode choice. To explain the shift in balance toward more cash payments, some of these determinants must have undergone a significant evolution. Identifying them would provide an ex post confirmation of their importance. Boone et al. (2014) seek to do so using a predictive model of payment mode choice built on many determinants from extant literature. Yet their model can only partially explain the observed time trend in M&A payment modes. Boone et al. also note that other variables could be correlated by coincidence with the evolution in payment modes and conclude that “propensities (to use mixed payments and stock) cannot be explained by our measures designed to capture traditional theories for the payment choice” (Boone et al. 2014, p. 297). Identifying the main driver of the marginalization of full stock payments in the United States thus remains an open issue.

Figure 1 delivers a second puzzling observation: The long-term decrease in full stock payments started in 2001 and the 1990–2000 subperiod displayed no clear trend. In 2000, 62% of transactions were fully paid in

Figure 1. (Color online) Percentage of Fully Stock-Paid M&A Transactions in the United States During 1990–2014



Notes. This figure displays the percentage of fully stock-paid M&A transactions in the United States from 1990 to 2014 (count and value based). The sample is collected from the Thomson SDC database, using the selection criteria listed in Section 2, which produced a set of 6,955 M&A transactions.

stock, but only 23% remained so in 2003. This decline then continued, but at a slower pace, the percentage of fully stock-paid transactions fluctuating around 10% from 2010 and after. We accordingly note that the Statements of Financial Accounting Standards (SFAS) 141 and 142 were adopted in June 2001. SFAS 141 dropped the pooling method of accounting for M&A transactions, imposing de facto, the purchase method application to all transactions. SFAS 142 abolished the goodwill amortization principle, replacing it by a yearly impairment test procedure. The pooling method of accounting, based on Accounting Principles Board (APB) Opinion 16 from 1970, consists of the simple addition of the income and balance sheet items to produce the new merged entity financial statements. The purchase method instead imposes a reevaluation of the target's assets and liabilities at fair value and the recognition of goodwill if a difference exists between the paid price and the reevaluated net assets. If they opted for pooling,² acquirers could avoid potential penalizing effects on their financial ratios due to their acquisitions.³ The synchronicity between the sharp drop in the evolution of fully stock-paid M&A transaction frequency and adoptions of SFAS 141 and 142 is questioning. Can these accounting rule changes help explain the marginalization of full stock payments in U.S. M&A transactions? The question is worthy of investigation not only because of the importance of the M&A mode of payment choice in itself but because the question of whether accounting rules matter for economic agents is still largely open to debate in the academic community (Amel-Zadeh et al. 2015).

Ali and Kravet (2016) contribute to this suspicion. The authors study the implications of pooling abolishment on two M&A outcomes: the mode of payment and the takeover probability. They focus on the role of the Step-Up (the difference between the purchase price and the target's book value of net assets) because this conditions the financial reporting implications of pooling. Their main finding concerning the mode of payment choice is to uncover a positive relation between the target Step-Up and full stock payment during the pooling period, which disappears after pooling abolishment. Ali and Kravet (2016) infer from this observation that pooling had driven full stock payment before 2001. The authors also suggest that pooling abolishment has decreased the probability of full stock payment since 2001, even if this trend is not observable in their particular sample. Other evidence of the importance of pooling as a determinant of the payment mode has been reported in the literature. In the specific case of AT&T's acquisition of NCR, Lys and Vincent (1995) note, for example, that AT&T decided to pay fully in stock to qualify for pooling.

The adoptions of SFAS 141 and 142 were one-time experiences and, around them, many synchronous events may have had an effect. Nearly any variable displaying a structural break after 2001 could appear to be a statistically significant determinant of the evolution in full stock payments.⁴ For example, in the wake of the September 11, 2001, attacks, the U.S. Federal Reserve drastically lowered its federal fund target rate to avoid an economic recession. This change in interest rate policy affected the cost of raising cash to finance M&A transactions. The bursting of the Internet bubble also was contemporaneous with SFAS 141 and 142 but also strongly affected the M&A market activity, especially in high-tech industries, and stock-paid transactions are known to be more frequent in high growth industries (Eckbo et al. 2017). Therefore, to test for a causal relation between pooling abolishment and the marginalization of the full stock payment in the United States, we need an identification strategy.

To apply one, we use Canada as a counterfactual for the United States as before 2001 pooling was also allowed under the Canadian Institute of Chartered Accountants (CICA) Handbook Section 1580.⁵ In 2001, as in the United States, this possibility was abolished under the CICA Handbook Section 1581. Moreover, the U.S. and Canadian economies are closely tied, so Canada is a prime candidate to serve as a counterfactual, as Eckbo (1992) suggested in his effort to identify a possible deterrence effect of M&A regulations. In support of our identification strategy, we note first that the U.S. and Canadian conditions for pooling, before its abolishment, differed substantially. In the Canadian case, under the CICA Handbook Section 1580, pooling was allowed only if one of the parties could not be

readily identified as the acquirer. This strict restriction led to very limited uses of pooling during the 1990s. Accordingly, the abolishment of pooling in Canada would likely have had, at best, a *very limited* impact on incentives to opt for stock as a payment medium. In this sense, pooling abolishment in Canada acts as a placebo for the “medication” of pooling abolishment in the United States. The contemporaneous abolishment of pooling in the United States and Canada contributes to our identification strategy. Common shocks to the U.S. and Canadian economies are controlled for using the difference-in-differences test that we implement, and therefore time synchronicity strengthens the control for these sources of latent factors. Recently, Cedergren et al. (2015) also use Canada as a counterfactual for the U.S. experience to control for endogeneity concerns in their study of the relation between goodwill amortization abolishment and acquisition profitability and risk.

For our baseline analyses, we collected two large M&A transaction samples from the Thomson SDC database for the 1990–2014 period: one for the United States and one for Canada.⁶ We used identical selection criteria: deal size above USD 1 million, public acquirers (no restriction on target status), a ratio of deal value to acquirer size of at least 1%, exchange offers, acquisition of assets, acquisition of certain assets, buybacks, recaps and acquisition (of stock) excluded, the percentage of shares acquired between 50% and 100%, 100% of shares held after the transaction, the consideration offered reported in the Thomson SDC database, and financial acquirers (Standard Industrial Classification (SIC) codes 6000–6999) excluded. The samples comprise 6,955 U.S. and 1,712 Canadian transactions.

We start by reporting stylized facts about payment mode choices over the 25-year period. As Figure 1 highlights, full stock payments declined abruptly after 2001 in the United States, but that was not the case in Canada, as confirmed by a Chow test of the structural break. After providing the descriptive statistics for a set of traditionally observed determinants of M&A payment choices, we estimate a linear probability model for stock-payment choices in the United States and Canada. The results confirm that the data sets are comparable to previous studies; we find similar historical results.

For the main analysis, we merge the U.S. and Canadian samples and use a difference-in-differences test, which is robust to many potential sources of bias (Roberts and Whited 2013). We identify pooling and post-pooling abolishment periods, then test whether full stock payments declined significantly more in the United States than in Canada after pooling abolishment. They did; this result is statistically highly significant, such that the probability of opting for full stock payments shrank by more than 20 percentage points (pp) (or 30 pp, depending on the difference-in-differences specification) in the United States during

the post-pooling period, compared with Canada. This result provides strong support to the thesis of pooling abolishment being a main driver of full-stock-payment marginalization in the United States.

Among the additional analyses we provide, we attempt to identify the channel through which pooling abolishment might have had such a significant impact on the M&A market. Healy (1985) points out that when executives are rewarded by earnings-based bonuses, they are more likely to select accrual policies and accounting procedures favorable to them. Therefore, we anticipate that pooling abolishment should have had a stronger influence on payment choices when CEO incentives are closely tied to financial performance (as CEOs are key decision makers in M&A processes; Harding and Rovit 2004). Our results support this prediction, indicating that reported financial performance matters for CEOs and suggesting one channel through which pooling abolishment has affected the M&A mode of payment choice. In contrast, we find that the combined evolution of classic determinants fails to explain the evolution of full stock payments, according to a comparison of their predictive power between the United States and Canada in the post-pooling period. This result corroborates the mixed results reported by Boone et al. (2014).

We also study whether, in the United States, the acquirers who selected full stock payments changed between the pooling and post-pooling periods. We find that after 2001, acquirers paying in stock are smaller and more leveraged, own more tangible assets, distribute dividends more often, more frequently enter into international transactions, and target more specifically public firms. By identifying the real change in acquirers' profiles and the characteristics of full-stock-payment transactions after the abolishment of pooling, we corroborate the material impacts of this abolishment on the M&A market.

As another complementary investigation, we test whether U.S. deals that probably would have used pooling, if it were still allowed, were paid in stock during the post-pooling period. They were not, thus providing further evidence that pooling represented a motivation to pay in stock during the 1990s, a result consistent with Ali and Kravet (2016).

We then address the value consequences of the abolishment for U.S. acquirers. We show that acquirers' cumulative abnormal returns (CAR) around the announcement of a stock-paid M&A transaction fell by 4.63 pp for public targets in the post-pooling period compared with the pooling period. This value effect is highly significant, both economically and statistically.

We finally provide the results of a long list of robustness checks for the baseline analysis, such as excluding high-technology firms to control for the Internet bubble, excluding cross-border deals, or using Europe

as a counterfactual instead of Canada (among others). As we detail subsequently, all the robustness checks confirmed the baseline results.

With these findings, this article contributes to the M&A payment choice literature. We first document the sharp decrease in full stock payment since 2001. We then show that the determinants suggested in traditional finance literature are poor at explaining this evolution. Pooling abolishment instead appears to be a first-order factor. This result contributes to the financial regulation literature by showing how far-reaching a change in accounting principles can be for the M&A market. Our results also contribute to the CEO compensation contract literature by pinpointing how incentives shape behavior. These insights are of first-order importance considering the weight of the M&A market as a channel for resource allocations (Andrade et al. 2001).

2. Data

Our baseline M&A data for both the U.S. and Canadian data⁷ came from the Thomson SDC database, with a set of selection criteria similar to those used by Betton et al. (2008):

- Deal size greater than USD 1 million;
- Public acquirers (but no restriction on target status);
- Ratio of the deal value to the acquirer size of at least 1%;
- Exchange offers, acquisition of assets, acquisition of certain assets, buybacks, recaps, and acquisition (of stock) excluded;
- Percentage of shares acquired between 50% and 100%;
- 100% of shares held after the transaction;
- The consideration offered is reported in the Thomson SDC database;
- Financial acquirers (SIC codes 6000–6999) excluded.

Table 1 contains the number and aggregate value (2010 constant USD) of transactions by year in the U.S. and Canadian samples. The wave of M&As at the end of the 1990s is clearly apparent in the U.S. sample, particularly for value-based statistics. That period witnessed gigantic, wealth-destroying transactions (Moeller et al. 2005). The rebirth of M&A market activity around 2004–2006 appears in both samples, though with some lag for the Canadian case. The Canadian sample was tiny in the early 1990s, reflecting our relative deal size selection restriction. Computing the relative size requires collecting the acquirer's market value, which is difficult in the Canadian sample for transactions in the early 1990s. André et al. (2004) cite the same challenge. Because we used the Canadian pooling and post-subsamples as the control group in our difference-in-differences test, with 266 transactions

Table 1. M&A Sample

Year	United States		Canada	
	Number of deals	Value of deals	Number of deals	Value of deals
1990	86	10,679	4	93
1991	141	17,044	4	135
1992	188	22,208	5	863
1993	226	23,133	8	801
1994	296	43,621	13	1,052
1995	396	54,256	24	3,223
1996	431	69,443	35	4,195
1997	511	105,110	30	2,913
1998	536	90,868	23	3,123
1999	476	104,495	36	2,918
2000	482	94,601	41	4,266
2001	258	40,704	43	6,570
2002	237	35,501	55	5,155
2003	236	34,623	70	4,971
2004	284	40,858	94	8,890
2005	322	45,540	122	6,876
2006	273	45,345	149	13,100
2007	276	46,672	169	15,249
2008	203	27,656	138	11,434
2009	175	23,034	143	7,848
2010	195	34,473	123	10,666
2011	191	28,627	122	10,731
2012	165	26,713	92	7,549
2013	178	29,197	74	5,269
2014	193	28,273	95	8,917
Total	6,955	1,122,675	1,712	146,807

Notes. This table presents the M&A samples for the United States and Canada. The Thomson SDC database is our data source. The sample selection criteria are: deal size above USD 1 million, public acquirers (no restriction on target status), deal value to acquirer size of at least 1%, exchange offers, acquisition of assets, acquisition of certain assets, buybacks, recaps and acquisition (of stock) excluded, percentage of shares acquired between 50% and 100%, 100% of shares hold after transaction, consideration offered reported by the Thomson SDC database, and financial acquirers (SIC codes 6000–6999) excluded. Values are reported in 2010 constant million USD for comparability through time and between the United States and Canada.

between 1990 and 2001 and 1,446 transactions between 2001 and 2014, we have sufficient observations. Moreover, the relatively smaller sample size, if anything, reduces the statistical power of the analyses, thus offering a more conservative test.

We collected market data from the Center for Research in Security Prices (CRSP) database for the United States and Datastream for Canada. If the prices or number of shares required to compute the market values were unavailable in these databases, we collected market values from the Thomson SDC database (AMV field). We used the Compustat Merged database for U.S. firms' financial statements and the Compustat North America database for Canadian ones. Macroeconomic information (interest rates, consumer price

index, credit spread) were from the Federal Reserve Bank of St. Louis's Economics Data website.⁸

3. Stylized Facts About the M&A Method of Payment

3.1. Evolution of Full Stock Payments Over Time

Table 2 contains the time series of fully stock-paid M&A transaction percentages in the United States and Canada, in value (2010 constant USD) and count. As observed in Figure 1, the U.S. time series exhibits a sharp decline in 2001 and 2002, which then continues at a reduced pace. No such evolution appears in Canada (yearly average stock-paid transaction percentages in value are 37.22% during 1990–2001 and 37.21% during 2002–2014; the corresponding count-based percentages are 47.01% and 46.48%). To confirm the significantly different evolutions between the United States and Canada, we implement a Chow test of structural break for both countries:

$$\text{stock Paid } \%_t = \alpha + \beta \text{Post}_{\text{pool}_t} + \gamma \text{Trend}_t + \delta(\text{Trend}_t \times \text{Post}_{\text{pool}_t}) + \varepsilon_t \quad (1)$$

where *Stock Paid* $\%_t$ is the fully stock-paid transaction percentage in year t , $\text{Post}_{\text{pool}_t}$ is a dummy variable equal to 1 during the post-period (2002 onward), Trend_t is a linear time trend variable, and $\text{Trend}_t \times \text{Post}_{\text{pool}_t}$ is the interaction between the linear time trend and the $\text{Post}_{\text{pool}_t}$ dummy variable. We test for the presence of a structural break in 2001 with the joint test of significance $\beta = \delta = 0$.

For the U.S. time series, the β coefficient is negative (significant, $p = 0.07$), and the Fisher statistic Chow test of the structural break is 12.44 (highly significant, $p = 0.00$), confirming the presence of a structural break in 2001. The Canadian experience differs fundamentally: The β coefficient is positive but not significant, and the Fisher statistic is 0.27 (not significant, $p = 0.76$), so there was no structural break.¹⁰ In the U.S. data, the interaction term δ coefficient is not statistically significant either. That is to say, the structural break is driven by a change in level, not slope.

3.2. Determinants of Stock Payment

We used a large set of determinants of the M&A mode of payment, as identified by prior M&A literature (Eckbo et al. 2017): deal size (USD million), acquirer size (USD million), cash holding, market-to-book ratio, asset tangibility, research and development (R&D), dividend payments, leverage ratio, target status, relative size, horizontal deal, domestic deal, and the 10-year Treasury bond interest rate. All the variable definitions are in Table A.1 in the appendix.

Table 3, panel A reports the descriptive statistics for the United States (panel A.1) and Canada (panel A.2). The first three columns refer to the whole period,

Table 2. Fully Stock-Paid Transaction Percentages by Year and Structural Break Test

Panel A: Stock-paid transaction percentages by year				
Year	United States		Canada	
	% stock paid (value)	% stock paid (count)	% stock paid (value)	% stock paid (count)
1990	68.98	50.00	58.07	50.00
1991	46.51	48.23	95.07	75.00
1992	48.54	54.79	11.91	80.00
1993	40.02	46.02	72.43	37.50
1994	48.47	51.35	26.46	38.46
1995	58.10	55.81	7.44	37.50
1996	52.71	58.93	30.96	37.14
1997	43.74	53.82	32.13	53.33
1998	48.23	49.07	18.98	43.48
1999	52.92	57.14	28.23	33.33
2000	61.90	56.85	22.85	34.15
2001	39.45	41.86	42.15	44.19
2002	25.06	33.76	38.16	38.18
2003	22.66	32.20	35.02	44.29
2004	19.99	25.35	36.41	53.19
2005	18.69	24.53	64.06	50.00
2006	14.23	20.51	49.70	41.61
2007	12.64	21.38	35.17	40.24
2008	14.81	27.59	58.48	47.83
2009	14.97	30.86	38.94	59.44
2010	14.78	25.13	22.62	46.34
2011	7.55	20.94	26.74	39.34
2012	8.17	19.39	16.08	46.74
2013	5.97	19.66	28.46	48.65
2014	10.91	21.76	33.94	48.42
Grand average	35.80	41.22	37.18	45.79

Panel B: Structural break test				
	United States		Canada	
	Coeff.	p-value	Coeff.	p-value
$\text{Post}_{\text{pool}} \times \text{Trend}$	−0.0093	(0.12)	0.0010	(0.88)
$\text{Post}_{\text{pool}}$	−0.1073	(0.07)	0.0251	(0.77)
Trend	0.0011	(0.84)	0.0020	(0.72)
Constant	0.5130	(0.00)	0.3832	(0.00)
R^2	91.24%		25.87%	
No. of observations	25		22	
Chow test	12.44	(0.00)	0.27	(0.76)

Notes. Panel A presents the time series of fully stock-paid M&A transaction percentages in the United States and Canada, from 1990 to 2014, in value (2010 USD constant) and count. The grand average is the average across the sample of M&A transactions. Panel B displays a Chow test of the structural break with a known change of structure date (estimated for 1990–2014 for the United States and 1993–2014 for Canada). *Trend* is a linear time trend variable, and *Post* is a dummy variable that is equal to 1 for the post-pooling abolishment period. The Chow test is the Fisher statistics of a joint test of significance on the $\text{Post}_{\text{pool}}$ dummy variable coefficient and its interaction with the linear time trend coefficient.

from 1990 to 2014, and include the mean, median, and standard deviation. The next two columns display the means by pre- and post-pooling periods; finally, the last column provides the p -value results of a classic

Table 3. Descriptive Statistics

Panel A.1: United States						
	1990–2014			1990–2001	2002–2014	Diff. avg.
	Mean	Median	Std. dev.	Mean	Mean	p-value
<i>Deal Size</i> (million USD)	130	128	33	111	147	(0.01)
<i>Acquirer Size</i> (million USD)	1,624	1,504	749	1,236	1,983	(0.01)
<i>Acquirer Cash</i>	16.36%	16.66%	2.30%	15.19%	17.45%	(0.01)
<i>Acquirer Market to Book</i>	2.59	2.38	1.02	3.05	2.16	(0.03)
<i>Acquirer Tangibility</i>	38.95%	38.63%	4.60%	41.12%	36.94%	(0.02)
<i>Acquirer Research and Development</i>	6.15%	5.63%	1.59%	5.83%	6.44%	(0.34)
<i>Acquirer Dividend</i>	24.34%	23.31%	6.83%	24.41%	24.28%	(0.96)
<i>Acquirer Leverage</i>	15.08%	14.97%	1.40%	14.75%	15.40%	(0.25)
<i>Public Target</i>	30.45%	29.71%	8.19%	35.29%	25.99%	(0.00)
<i>Private Target</i>	54.92%	55.58%	6.37%	52.69%	56.98%	(0.10)
<i>Relative Size</i>	0.55	0.53	0.16	0.46	0.64	(0.00)
<i>Horizontal</i>	35.76%	35.15%	3.33%	33.82%	37.55%	(0.00)
<i>Domestic</i>	85.49%	85.07%	5.59%	90.19%	81.14%	(0.00)
<i>10-Year Interest Rate</i>	6.33%	6.51%	1.49%	7.63%	5.13%	(0.00)
Panel A.2: Canada						
	1990–2014			1990–2001	2002–2014	Diff. avg.
	Mean	Median	Std. dev.	Mean	Mean	p-value
<i>Deal Size</i> (million USD)	76	82	25	74	78	(0.71)
<i>Acquirer Size</i> (million USD)	483	400	313	310	643	(0.00)
<i>Acquirer Cash</i>	11.11%	10.91%	5.85%	6.59%	15.29%	(0.00)
<i>Acquirer Market to Book</i>	1.88	1.64	0.80	1.66	2.09	(0.18)
<i>Acquirer Tangibility</i>	77.16%	68.72%	21.93%	87.13%	67.97%	(0.03)
<i>Acquirer Research and Development</i>	1.75%	1.36%	1.71%	1.07%	2.38%	(0.05)
<i>Acquirer Dividend</i>	21.56%	21.21%	11.80%	21.14%	21.95%	(0.87)
<i>Acquirer Leverage</i>	14.45%	13.36%	7.36%	19.81%	9.51%	(0.00)
<i>Public Target</i>	49.30%	47.30%	14.31%	59.34%	40.03%	(0.00)
<i>Private Target</i>	38.15%	40.00%	13.64%	30.48%	45.23%	(0.01)
<i>Relative Size</i>	1.29	1.13	0.76	1.65	0.95	(0.02)
<i>Horizontal</i>	55.65%	55.80%	11.02%	59.00%	52.56%	(0.16)
<i>Domestic</i>	76.42%	72.73%	12.59%	82.39%	70.91%	(0.02)
<i>10-Year Interest Rate</i>	5.36%	5.33%	2.33%	7.28%	3.60%	(0.00)
Panel B: United States—Complementary analyses						
Variable	Mean	Median	Std. dev.	N		
<i>% Variable Compensation</i>	71.78%	78.73%	23.19%	1,146		
<i>10-Year Interest Rate</i>	6.65%	6.81%	1.21%	1,157		
<i>Acquirer CAR</i>	1.59%	0.93%	11.70%	5,148		
<i>Acquirer Free Cash Flow</i>	−0.0197	0.0427	0.2274	5,148		
<i>Credit Spread</i>	2.65%	2.22%	1.45%	1,157		
<i>High Tech</i>	47.19%	0.00%	49.94%	1,157		
<i>Hostile</i>	0.47%	0.00%	6.81%	5,148		
<i>Manufacturing</i>	20.05%	0.00%	40.06%	1,157		
<i>Pooling Hat</i>	36.29%	34.49%	18.47%	1,157		
<i>Run Up</i>	1.1745	1.0248	0.6808	5,148		
<i>Sigma</i>	3.83%	3.27%	2.30%	5,148		
<i>Step Up</i>	3.19	1.88	5.29	1,157		
<i>Stock</i>	37.60%	0.00%	48.44%	5,148		
<i>Target Leverage</i>	15.24%	5.84%	19.58%	1,157		
<i>Target Return on Assets</i>	−2.02%	4.46%	23.43%	1,157		
<i>Target Size</i> (million USD)	168	102	185	1,157		
<i>Tender Offer</i>	8.76%	0.00%	28.27%	5,148		

Notes. Panel A reports the descriptive statistics for the set of payment mode determinants classically used in M&A literature (see Eckbo et al. 2017) that we used in our main analysis for the United States (panel A.1) and for Canada (panel A.2). The M&A samples for the United States and Canada are presented in Table 1. The pre-pooling abolishment period is 1990–2001, and the post-pooling period is 2002–2014. Statistics are computed on yearly averages. “Diff. avg.” is a standard test of difference of means. Panel B reports the descriptive statistics of the variables used in complementary analyses focusing on the U.S. sample only. Reported figures are for the largest available number of observations for each variable. All variables are defined in Table A.1.

test of the difference in means across the two subperiods. The U.S. sample differs from the one that Eckbo et al. (2017) use in that the covered periods are different. Eckbo et al. (2017) gather a sample of 4,919 U.S. transactions from the Thomson SDC database over 1980–2008, whereas we focus on the 1990–2014 period. Although the measures of the acquirer's market-to-book ratio (2.56 in our sample versus 3.06 in Eckbo et al.), leverage (15% versus 16.9%), and asset tangibility (38.9% versus 42.9%) are comparable, our acquirers are significantly smaller (USD 1,256 million in average total assets—unreported—versus USD 3,218 million), pay dividends less frequently (24% vs. 44%), and do more R&D (6.1% of total assets versus 4.4%). Concerning comparisons between the pre- and post-pooling periods, for both the United States and Canada, we find significant differences, which importantly raises the question of whether these evolutions can explain the observed decline in frequency of full stock payments during the post-pooling period. In the United States, the deal size, acquirer size, cash holdings, relative size, and frequency of horizontal transactions all increase. The acquirer's market-to-book ratio and asset tangibility both decrease, as do the frequency of public targets, domestic transactions, and 10-year Treasury bond interest rate levels. In Canada, the evolutions of the acquirer's size, cash holdings, asset tangibility, frequency of public targets, domestic deals, and 10-year Treasury bond interest rate are comparable to those for the United States. However, we observe three differences: an increase in acquirer R&D (absent in the United States), a decrease in acquirer leverage (absent in the United States), and a decrease in relative size (increasing in the United States). In panel B we report descriptive statistics for 20 additional variables used in complementary analyses with the U.S. sample. For each variable, the reported figures refer to the total number of available observations. These displayed statistics are of the same order of magnitude as reported in previous studies for comparable samples (see Betton et al. 2008, Harford and Li 2007).

For more insight into the comparability of our sample with extant samples, we also reproduce the classic multivariate probability of stock-payment analysis. Yet we opt for a linear probability model; most studies instead use a nonlinear model, because their dependent variable is binary. Our choice is motivated by our willingness to test the statistical significance of the coefficient variations between the pooling and post-pooling periods. This test can easily be implemented by merging observations from the two periods into one common sample and including interaction terms between the variables of interest and a $Post_{pool}$ dummy variable. However, the interaction term coefficients in nonlinear models are not marginal effects and therefore must be interpreted with care (Greene 2010). The linear probability model is immune to this issue.¹¹

Table 4. Determinants of the Probability of Full Stock Payment

	United States		Canada	
	Coeff.	p-value	Coeff.	p-value
<i>Acquirer Size</i>	−0.0057	(0.18)	−0.0259	(0.02)
<i>Relative Size</i>	0.0066	(0.48)	−0.0016	(0.90)
<i>Acquirer Leverage</i>	−0.2032	(0.00)	−0.2649	(0.03)
<i>Acquirer Market to Book</i>	0.0197	(0.00)	0.0226	(0.00)
<i>Acquirer Tangibility</i>	−0.0594	(0.01)	0.0105	(0.84)
<i>Acquirer Dividend</i>	−0.0605	(0.00)	−0.1340	(0.00)
<i>Acquirer Research and Development</i>	0.4451	(0.00)	0.6816	(0.00)
<i>Acquirer Cash</i>	−0.0393	(0.37)	−0.1023	(0.32)
<i>Domestic</i>	0.0787	(0.00)	0.1446	(0.00)
<i>Horizontal</i>	−0.0163	(0.21)	0.0551	(0.10)
<i>Public Target</i>	0.0670	(0.00)	0.2434	(0.00)
<i>10-Year Interest Rate</i>	11.4812	(0.00)	0.0870	(0.93)
Sector fixed effects	Yes		Yes	
R ² (%)	20.83		30.46	
No. of observations	5,337		786	

Notes. This table displays the results of a linear probability model of full stock payments using the set of mode of payment choice determinants classically used in prior literature (Eckbo et al. 2017). All variables are defined in Table A.1.

As the results in Table 4 indicate, the coefficient estimates for the U.S. sample are close to those reported by Eckbo et al. (2017): acquirer market-to-book ratio and R&D increase the probability of full stock payments; dividend payments and leverage decrease it. Acquirer size and asset tangibility retain the same signs, but the former loses its significance, and the latter was not significant in Eckbo et al.¹² Despite the compositional differences revealed in Table 3, our sample thus appears representative of classic samples used in prior M&A literature to study the determinants of payment choice. Finally, the comparison of the multivariate analyses for the United States and Canada indicates that, with the exception of acquirer tangibility, all the statistically significant U.S. coefficients retain their signs in the Canadian case (and mostly remain statistically significant). These results strengthen our decision to use Canada as a counterfactual for the U.S. pooling abolishment experience.

4. Pooling Interests vs. Purchase Accounting Methods

To test whether pooling abolishment drove the marginalization of full stock payments in the United States, we first summarize the relevant accounting regulations in both nations. This initial analysis is important to understand why Canada is a valid counterfactual, namely, because pooling was possible in Canada before 2001.

4.1. The U.S. Case

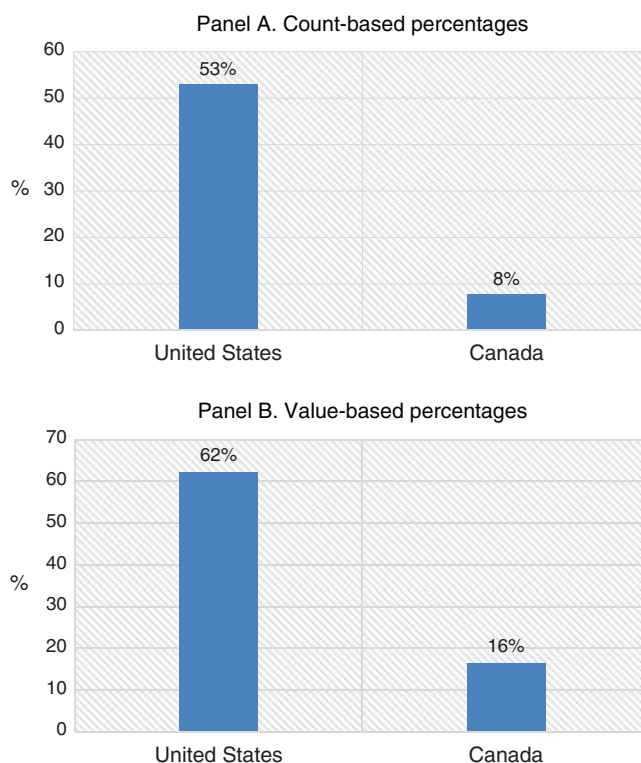
The SFAS 141 and 142 introduced two major reforms in 2001: abolishing the pooling of interests method¹³ and

goodwill amortization.¹⁴ Before 2001, M&A accounting methods followed APB Opinion 16, from 1970, which allowed two methods: pooling of interests and purchase. With the pooling method, a simple sum of the income and balance sheet statements of the two merging companies served as the input for the financial statements of the newly merged entity. This procedure avoided any asset revaluations or goodwill recognition. The purchase method instead used a fair-value reevaluation of target assets and liabilities before incorporating them into the acquirer's financial statements. If the acquisition price represented a premium with respect to the fair-value re-evaluation process, as was the case especially at the end of the nineties due to the increase in market valuations, a goodwill was recognized. Before 2001, this goodwill had to be amortized over its useful life, with a maximum of 40 years (APB Opinion 17). Another significant difference between pooling and purchase involved the day on which target net incomes were taken into account in the newly merged financial statements. Pooling required such consideration from the beginning of the fiscal year, whereas under the purchase method it began with the acquisition date.

Before 2001, the M&A accounting method was not, strictly speaking, a choice. Rather, APB Opinion 16 listed 12 criteria that, if met, allowed pooling. The merging parties structured their transactions to fulfill (or not) these criteria. The general idea was that pooling should apply to mergers of equals, so the main criteria were autonomy (i.e., merging companies could not be divisions or subsidiaries of one another in the two years before the merger), a single transaction (merger should be a one-step process, completed within a year of its initiation), and an essentially stock-for-stock transaction (at least 90% of the consideration).¹⁵ The method was common for large M&A transactions in the United States, as Figure 2 displays. It indicates the average percentage of M&A transactions using pooling with deal sizes greater than USD 100 million that were fully paid by stock during 1990–2001 (source: Thomson SDC database). Panel A displays count-based percentages, and panel B contains value-based percentages. Pooling was used for more than one out of two transactions. Reda (1999) reports that, in 1997, the dollar volume of pooling exceeded that of purchase by a factor of 20 in the United States. Thus, the pooling method was the method of choice for large, U.S., M&A transactions paid for fully by stock.

With this stylized fact, several researchers have compared the effects of pooling and purchase rules on financial ratios such as the EPS and ROA of the newly merged entity. The general view is that purchase accounting degrades them relative to pooling (see Reda 1999). But this depends on the market valuations and operating performance levels of the acquirer

Figure 2. (Color online) Pooling vs. Purchase in the United States and Canada During 1990–2001



Notes. This figure displays the average percentage of M&A transactions in our sample (Table 1) that used pooling and had a deal size greater than USD 100 million and were fully paid by stock during 1990–2001 (source: Thomson SDC database). Panel A reports the count-based percentage, and panel B reports the value-based percentage.

and the target as well as on the financial ratios under consideration.¹⁶ To gain a better understanding of this issue, one must disentangle changes in earnings (the numerator of financial ratios) from changes in total assets or number of shares outstanding (their denominators). On the numerator side, by imposing goodwill amortization, purchase accounting penalizes the performance. But during the pooling period, financial analysts and investors were mostly looking at operating performance excluding goodwill amortization. In such cases, effects of purchase accounting are limited. Moreover, adoptions of SFAS 141 and SFAS 142 were contemporaneous. Pooling was abolished concurrently with goodwill amortization (and its replacement by yearly impairment tests). Switching from the pre-SFAS 141 and SFAS 142 regime to the post one had, therefore, most probably a limited impact on the earnings side, at least as viewed by financial analysts and investors. On the denominator side, the situation depends on whether we focus on the number of shares outstanding or on total assets. Concerning the number of shares outstanding, full stock payment imposed by pooling mechanically leads to its increase.¹⁷ Switching from the

pooling period to the post-pooling periods released the full-stock-payment constraint and led to an increase in EPS if acquirers opted more frequently for cash or mixed payment modes. For financial ratios scaled by total assets such as ROA, the fair valuation process imposed by purchase accounting unambiguously leads to an increase in the denominator. Hence, pooling abolishment penalized reported ROA of newly merged entities. Anticipating the net effects of pooling abolishment on financial ratios reported by new merged entities is clearly subtler than it may appear at first sight. It depends on the financial ratio considered, the financial market valuation level, and the comparative situation of the acquirer and the target, both in terms of valuation and operating performances. Even the target past accounting practice matters because it impacts the fair valuation process imposed by purchase accounting.

Beyond the effects on reported financial ratios, SFAS 141 and SFAS 142 impose asset fair valuation and yearly impairment tests, which may have promoted transparency and efficiency of capital allocation in the M&A market. In particular, the yearly impairment test procedure imposed by SFAS 142 is a channel through which overpayment is revealed to shareholders. This mechanism put CEOs (and other M&A decision makers) under fire in case of overpayment by disclosing poor pricing decisions, imposing *de facto* a stronger market discipline in the M&A market. Consequently, adoption of SFAS 141 and 142 did not only drop pooling and goodwill amortization but also affected the M&A market resource allocation processes, a potential confounding factor.

Several researchers have considered the relation between pooling and the choice of payment methods, as well as managers' motivations to choose pooling and the value effect for shareholders. Ali and Kravet (2016) use the adoption of SFAS 141 and 142 as a kind of natural experiment to study the relation between accounting regulations, the financing used in M&A transactions, and the takeover probability. As noted previously, they focus on the role of the target Step-Up because this variable conditions the financial reporting implications of choosing pooling. They establish a positive relation between the Step-Up and the probability of stock-for-stock financing before 2001. However, this relation disappears after the abolishment of pooling and goodwill amortization, leading the authors to infer that pooling was an important driver of stock-for-stock financing choices during the pooling period. The authors also suggest that pooling abolishment has decreased the probability of full stock payment since 2001, but they do not provide supportive evidence. The frequency of full stock payment does not decrease significantly in their sample after 2001 (Ali and Kravet 2016, Table 1),¹⁸ and also their logit regressions incorporate yearly fixed-effects, hiding any trend regarding the frequency of

fully stock-paid transactions. Moreover, in the absence of a clear identification strategy, inferring a causal effect between pooling abolishment and the probability of full stock payment is open to the classic omitted variables bias. Ayers et al. (2002) confirm that firms using pooling were ready to pay higher acquisition premia. Weber (2004) also studies market reactions to the Securities and Exchange Commission's (SEC) adoption of SAB 96, a new regulation that forced firms to choose between pooling and undertaking share repurchase programs in the two years following an acquisition. Most firms with pending pooling mergers at the time of SAB 96 adoption maintained pooling as an accounting method, at the cost of renouncing share repurchase programs. Therefore, pooling appears to have a real cost to shareholders. Aboody et al. (2000) further report that pooling was more likely when managers received earnings-based compensation, raising the issue of conflicts of interest with shareholders. With a sample of 324 U.S. stock swap acquisitions between 1990 and 1998, Martinez-Jerez (2008) reports a negative, statistically significant difference in market reactions to pooling versus purchase transactions, of approximately -4 pp, though the differential grows to -8 pp for firms with ineffective corporate governance. The authors conclude that investors interpret the choice of a purchase accounting method as a signal of management's confidence in the success of the transaction.

4.2. The Canadian Case

Pooling was abolished concurrently with goodwill amortization: the transaction had to be accomplished by an exchange of voting shares, and it had to be impossible to identify one of the combining firms as the acquirer.¹⁹ Because the merger partner with more than 50% of the combined firm is assumed to be the acquirer, practically speaking, pooling might seem to work only for combinations of firms of equal size. But Canadian Generally Accepted Accounting Principles (GAAP) also had other rules to determine the status of "acquirer" (CICA Handbook Section 1581); e.g., the entity that transfers cash or other assets or incurs liabilities, the entity that issues equity interests, the relative voting rights in the combined entity after the business combination, the existence of a large non-controlling interest in the combined entity if no other owners hold the largest noncontrolling voting interest in the combined entity, the composition of the governing body or the senior management of the newly merged entity, the terms of exchange of equity interests and the entity that initiated the transaction. Given so many criteria, an acquirer could usually be identified, and pooling was uncommon. According to André et al. (2004), among 267 transactions undertaken during 1980–2000, only 8 (3% of the sample) relied on pooling. Figure 2 confirms this evidence for the 1990–2001 period, for a sample limited to transactions with a

deal size greater than USD 100 million and paid fully by stock: pooling was seven times (four times) more frequent in the United States than in Canada according to count- (value-) based percentages. Then, the pooling method was abolished in 2001 under CICA Handbook Section 1581.

The Canadian experience is particularly interesting with respect to our research question. The pooling abolishment year is the same as that in the United States, and the nations are close, with closely tied economic environments (Eckbo 1992, Cedergrén et al. 2015). The very restrictive pooling usage conditions in Canada meant that this approach had, at most, a very limited material impact on the payment choice by construction: pooling abolishment in the Canadian case plays the role of a placebo in our research design, like in medical experimentation. The time synchronicity strengthens the ability of our difference-in-differences test to control for common shocks affecting the U.S. and Canadian economies simultaneously. These arguments motivate our choice of Canada as a counterfactual for the U.S. experience.

5. Full Stock Payment and Pooling Accounting Abolishment

5.1. Difference-in-Differences Tests

In this section, we explicitly test whether pooling abolishment is a valid candidate for explaining the evolution in full stock payments in the United States. The difference-in-differences test, with Canada as a counterfactual, should be robust to misspecification-based sources of bias, including endogenous missing variables (Roberts and Whited 2013). As argued in Section 4.2, business combinations were infrequently accounted for under pooling in Canada before pooling abolishment, and this allows us to use Canada as a placebo with respect to the U.S. case. Moreover, the simultaneous abolishment of pooling in the United States and Canada coupled with a difference-in-differences test controls for shocks common to both economies. Accordingly, we adopt two difference-in-differences specifications (Greene 2011):

$$\begin{aligned} Stock_i = & \alpha + \beta US_i + \gamma Post_{pool_i} + \delta(US_i \times Post_{pool_i}) \\ & + \theta'(\text{Sector FE}_i) + \varphi'(\text{Controls}_i) \\ & + \varepsilon_i, \quad \text{and} \end{aligned} \quad (2)$$

$$\begin{aligned} Stock_i = & \alpha + \beta US_i + \delta(US_i \times Post_{pool_i}) + \gamma(\text{Year FE}_i) \\ & + \theta'(\text{Sector FE}_i) + \varphi'(\text{Controls}_i) + \varepsilon_i, \end{aligned} \quad (3)$$

where i is the deal index, $Stock_i$ is a dummy variable equal to 1 for a full stock payment for deal i , US_i is a dummy variable equal to 1 if the acquirer is a U.S. firm but 0 if the acquirer is a Canadian firm, $Post_{pool_i}$ is dummy variable equal to 1 if the deal i announcement date is during the post-period (after June 30, 2001),

Sector FE_{*i*} is a vector of sector fixed effects (defined at the SIC two-digit level), **Year FE_{*i*}** is a vector of year fixed effects, and **Controls_{*i*}** is a vector of control variables. We use bold notation to identify vectors. The set of control variables is the same as in Table 4.

Our two difference-in-differences specifications are linear probability models (LPM), which are generally less well suited to the analysis of binary dependent variables than probit or logit specifications. However, we selected this estimator because Equations (2) and (3) both incorporate interaction terms. Coefficients of interaction terms in nonlinear models cannot be interpreted as marginal effects (Greene 2010), but in a linear specification, they are. Adopting LPM specifications, therefore, facilitates the interpretation of our results. The Equation (2) specification is a classic difference-in-differences test implementation in a multivariate context. We also report the results obtained with the Equation (3) specification, that corresponds to the logit specification used in Ali and Kravet (2016), because it includes year fixed effects controls for time-varying common factors that are not explicitly included in the vector of control variables (e.g., macroeconomic variables). The $Post_{pool_i}$ dummy cannot be included in this second specification though, because it is a linear combination of the set of year fixed effects.

In Table 5 we report the results from estimating Equations (2) and (3) in the first two columns. The M&A sample is the one we introduced in Section 2, though restricted to transactions for which all the required data fields are available. All control variables are as defined in Table A.1. In Figure 3 we display the evolution of full stock-payment frequencies in the United States (panel A) and Canada (panel B) as the red continuous line. The parallel trends assumption underlying a valid difference-in-differences test implementation is clearly respected, except for years at the very beginning of the sample period, for which the Canadian sample size is limited (see Table 1).

The test of the pooling abolishment hypothesis relied on the interaction term $US_i \times Post_{pool_i}$ with coefficient δ for Equations (2) and (3). In Table 5, column (1), the coefficient value is -0.3134 ($p = 0.00$), and in Column 2 it is -0.2099 ($p = 0.00$). These estimates strongly support the pooling abolishment hypothesis: In contrast with the Canadian experience, full stock payments declined in the United States in the wake of pooling abolishment. The result is robust to our introduction of a large set of determinants of the M&A mode of payment, industry-level latent factors that were constant over time, and annual common latent factors. The properties of the difference-in-differences specification also make it robust to any latent factors common to the United States and Canada. Moreover, the sizable coefficient values indicated a decline of 20 to 30 pp in

Table 5. Method of Payment and Pooling Abolishment: Difference-in-Differences Test

	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
US	0.1568	(0.00)	0.1183	(0.00)
$Post_{pool}$	0.0198	(0.66)		
$US \times Post_{pool}$	−0.3134	(0.00)	−0.2099	(0.00)
Acquirer Size	0.0048	(0.51)	0.0049	(0.51)
Relative Size	−0.0053	(0.17)	−0.0037	(0.35)
Acquirer Leverage	−0.2182	(0.00)	−0.2161	(0.00)
Acquirer Market to Book	0.0173	(0.00)	0.0191	(0.00)
Acquirer Tangibility	−0.0469	(0.03)	−0.0485	(0.03)
Acquirer Dividend	−0.0826	(0.00)	−0.0827	(0.00)
Acquirer Research and Development	0.4507	(0.00)	0.4369	(0.00)
Acquirer Cash	−0.0157	(0.69)	−0.0146	(0.71)
Domestic	0.1013	(0.00)	0.0993	(0.00)
Horizontal	−0.0046	(0.70)	−0.0065	(0.58)
Public Target	0.0853	(0.00)	0.0820	(0.00)
10-Year Interest Rate	2.2224	(0.00)	−1.0835	(0.49)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
R ² (%)	23.46		24.19	
No. of observations	6,123		6,123	

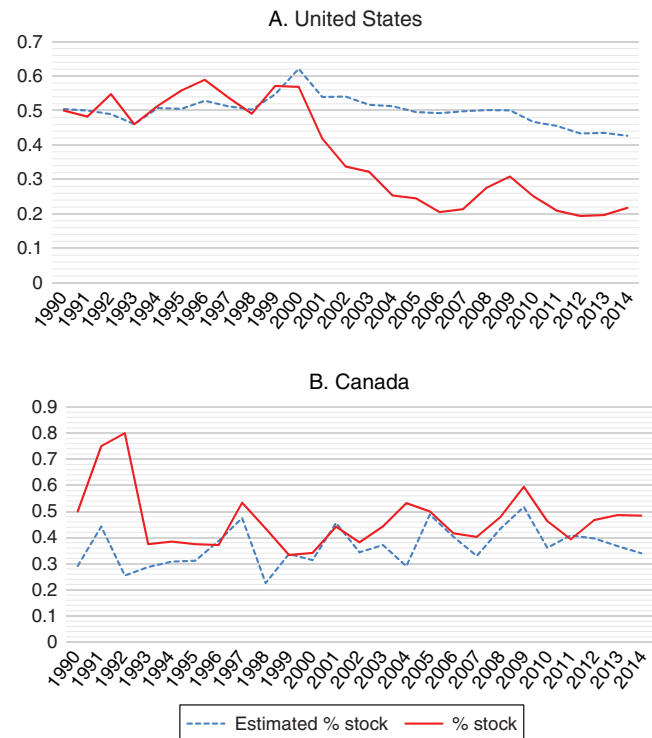
Notes. This table reports the results of two difference-in-differences test specifications. The dependent variable is the full-stock-payment dummy variable (equal to 1 if the transaction is fully paid in stock). The M&A sample for the United States and Canada is introduced in Table 1. All variables are defined in Table A.1. Both specifications rely on a linear probability model. In column (1) the $Post$ dummy variable (equal to 1 for the post-pooling abolishment period) is explicitly introduced. In column (2) we introduce year fixed effects. $US \times Post_{pool}$ is the interaction term between the US and $Post_{pool}$ dummy variables. The set of control variables is the same as in Table 4. Standard errors are robust to heteroskedasticity.

the probability of a full stock payment during the post-period.²⁰ Finally, among the control variables, in comparison with the results obtained for the United States during 1990–2014 (Table 4), the coefficients of acquirer leverage, market-to-book ratio, tangibility, dividend, R&D, domestic transaction, public target, and 10-year interest rate kept their signs and statistical significance. Results reported in column (2) are particularly interesting with respect to the potentially confounding role of the pooling abolishment market discipline effect highlighted in Section 4.1. Year fixed effects absorb any latent factors affecting the M&A market as a whole.²¹

6. Additional Evidence

6.1. CEO Incentives and Full-Stock-Payment Probability

Pooling and purchase methods of accounting have significantly different consequences for the new merged entity, in terms of its displayed financial ratios (see Section 4.1). Executives whose compensation packages depended on financial performance indicators such as EPS or ROA had clear incentives to opt for full stock payments during the pooling period. Healy (1985)

Figure 3. (Color online) Forecasted Frequency of Full Stock Payments in the United States and Canada

Notes. This figure represents the evolution of the percentage of M&A transactions fully paid in stock from 1990 to 2014 (panel A for the United States, panel B for Canada). The corresponding M&A samples are in Table 1. “% Stock” refers to the observed percentage. “Estimated % Stock” is obtained using the fitted probabilities from the linear probability models in Table 7, column 1990–2001.

already points out that when executives are rewarded by earnings-based bonuses, they are more likely to adopt accounting rules that are more favorable to them. Consistently, Aboody et al. (2000) report that pooling was more likely in large target Step-Up transactions when managers received earnings-based compensation. CEO incentives, therefore, might provide a channel to explain the interaction between pooling and the choice of payment methods in M&A transactions.

To study the role of CEO incentives, we compute the percentage of CEO variable compensation using the Execucomp database. We next reproduce the LPM of the full stock payment in the United States (Table 4) but adding this time the percentage of variable CEO compensation and its interaction with the $Post_{pool}$ dummy variable.²² The sample size decreased drastically (from 5,337 observations to 1,146), due to the limited data availability in the Execucomp database.

Table 6, panel A displays the estimation results. We report four specifications, depending on whether the independent variables of interest and the year and sector fixed effects are included. The two independent variables of interest are % Variable Compensation and its interaction with the $Post_{pool}$ dummy variable.

Table 6. CEO Incentives and the Probability of Full Stock Payment

Panel A: Probability of full stock payment—All sample								
	(1)		(2)		(3)		(4)	
	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value
$Post_{pool}$	−0.4358	(0.00)	−0.2382	(0.01)	−0.2598	(0.00)		
% Variable Compensation			0.1012	(0.20)	0.0839	(0.31)	0.0943	(0.23)
% Variable Compensation $\times$ $Post_{pool}$			−0.2863	(0.01)	−0.2625	(0.02)	−0.2748	(0.01)
Acquirer Size	−0.0004	(0.96)	−0.0327	(0.00)	−0.0401	(0.00)	−0.0335	(0.01)
Relative Size	−0.0337	(0.00)	0.0043	(0.67)	−0.0001	(0.99)	0.0083	(0.42)
Acquirer Leverage	−0.1680	(0.01)	−0.1575	(0.02)	−0.1534	(0.05)	−0.1247	(0.11)
Acquirer Market to Book	0.0250	(0.00)	0.0252	(0.00)	0.0255	(0.00)	0.0257	(0.00)
Acquirer Tangibility	−0.0049	(0.89)	−0.0012	(0.97)	−0.0156	(0.73)	−0.0047	(0.92)
Acquirer Dividend	−0.0597	(0.01)	−0.0619	(0.01)	−0.0362	(0.14)	−0.0541	(0.03)
Acquirer Research and Development	0.6010	(0.00)	0.5654	(0.00)	0.5013	(0.01)	0.4483	(0.02)
Acquirer Cash	−0.0536	(0.55)	−0.0336	(0.71)	−0.0815	(0.38)	−0.0467	(0.61)
Domestic	0.0673	(0.00)	0.0618	(0.01)	0.0709	(0.00)	0.0679	(0.01)
Horizontal	0.0170	(0.44)	0.0135	(0.53)	0.0022	(0.92)	0.0070	(0.76)
Public Target	0.0019	(0.93)	0.0020	(0.93)	0.0055	(0.81)	0.0003	(0.99)
10-Year Interest Rate	1.3119	(0.33)	1.0624	(0.43)	0.8766	(0.54)	−8.1663	(0.08)
Credit Spread	1.4140	(0.01)	1.4986	(0.01)	1.5976	(0.01)	5.0953	(0.08)
Year fixed effects	No		No		No		Yes	
Sector fixed effects	No		No		Yes		Yes	
R ² (%)	36.62		36.84		39.63		42.19	
No. of observations	1,146		1,146		1,146		1,146	
F-test	—	—	5.78	(0.02)	4.74	(0.03)	5.82	(0.02)
Panel B: Probability of full stock payment—High/low ROA and high/low MTB subsamples								
	(1)		(2)		(3)		(4)	
	High ROA		Low ROA		High MTB		Low MTB	
	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value
% Variable Compensation	0.1957	(0.11)	−0.0142	(0.90)	0.0893	(0.37)	0.0355	(0.79)
% Variable Compensation $\times$ $Post_{pool}$	−0.2923	(0.04)	−0.2675	(0.06)	−0.2531	(0.04)	−0.2271	(0.19)
Acquirer Size	0.0044	(0.76)	0.0198	(0.20)	0.0092	(0.46)	0.0046	(0.79)
Relative Size	−0.0160	(0.85)	−0.0308	(0.06)	−0.0567	(0.63)	−0.0268	(0.09)
Acquirer Leverage	−0.2248	(0.05)	−0.0583	(0.64)	−0.0310	(0.79)	−0.0537	(0.66)
Acquirer Market to Book	0.0375	(0.00)	0.0127	(0.08)	0.0088	(0.13)	−0.0232	(0.71)
Acquirer Tangibility	−0.0636	(0.28)	0.0429	(0.54)	0.0058	(0.92)	0.0350	(0.62)
Acquirer Dividend	−0.0340	(0.29)	−0.0432	(0.32)	−0.0200	(0.54)	−0.0785	(0.05)
Acquirer Research and Development	0.2963	(0.33)	0.4646	(0.07)	0.3674	(0.05)	0.2636	(0.58)
Acquirer Cash	−0.0963	(0.46)	0.0174	(0.89)	−0.1456	(0.14)	0.2345	(0.23)
Domestic	0.0308	(0.42)	0.1031	(0.01)	0.0636	(0.04)	0.0689	(0.07)
Horizontal	0.0122	(0.72)	0.0123	(0.73)	−0.0140	(0.64)	0.0425	(0.25)
Public Target	−0.0415	(0.24)	0.0335	(0.34)	0.0296	(0.33)	0.0143	(0.70)
10-Year Interest Rate	−12.0159	(0.05)	−5.0140	(0.47)	−13.4634	(0.03)	−3.0293	(0.67)
Credit Spread	3.9745	(0.30)	5.6257	(0.23)	5.6392	(0.16)	5.1211	(0.27)
Year fixed effects	Yes		Yes		Yes		Yes	
Sector fixed effects	Yes		Yes		Yes		Yes	
R ² (%)	53.50		38.79		64.70		26.33	
No. of observations	567		579		565		581	
F-test	1.04	(0.31)	6.55	(0.01)	2.87	(0.09)	2.06	(0.15)

Notes. Panel A displays the results of a linear probability model of full-stock-payment determinants. The variables of interest are the proportion of CEO variable compensation (% variable compensation) and its interaction with $Post_{pool}$ (dummy variable = 1 for the post-pooling abolishment period). All variables are defined in Table A.1. Standard errors are robust to heteroskedasticity. The F-test is the Fisher statistics of a test of significance for the sum of the coefficients % Variable Compensation and % Variable Compensation $\times$ $Post_{pool}$. Panel B displays results obtained using panel A, column (4) specification for subsamples build on Acquirer Return on Assets and Market to Book Ratios. High (low) ROA corresponds to the subsample of acquirers with return on assets above (below) the sample median return on assets. The same criteria apply to high (low) MTB, using the acquirer Market to Book ratio. Sample medians are recomputed each year.

The former is positive but not significant in all specifications (column (2), 0.1012, $p = 0.20$; column (3), 0.0839, $p = 0.31$; column (4), 0.0943, $p = 0.23$). In these same columns, the latter coefficient was negative and highly significant (-0.2863 , $p = 0.01$; -0.2625 , $p = 0.02$; -0.2748 , $p = 0.01$). The addition of % *Variable Compensation* and its interaction with the $Post_{pool}$ coefficient provides us the net effect of % *Variable Compensation* during the post-pooling period. We obtain a negative value that is significantly different from 0 (Fisher statistics: 5.78, $p = 0.02$; 4.74, $p = 0.03$; 5.83, $p = 0.02$). This suggests that when CEOs of acquirers earn a high proportion of variable compensation, they were less likely to use full stock payments after pooling abolishment. These results suggest that CEO incentives help explain the interaction between pooling abolishment and the marginalization of full stock payments in the United States. When we compare the column (1) results with those in columns (2) and (3), we find that half of the post-pooling abolishment effect was due to CEO incentives. That is, in column (1) the $Post_{pool}$ coefficient is -0.4358 , and in columns (2) and (3) it shrinks to -0.2382 and -0.259 respectively, or roughly half.

In Table 6, panel B we report results corresponding to Table 6, panel A, column (4) (specification with year and sector fixed effects) by subsamples of high and low acquirer return on assets (columns (1) and (2)) and high and low acquirer market to book ratios (columns (3) and (4)). High (low) ROA corresponds to the subsample of acquirers with return on assets above (below) the sample median return on assets. The same criteria apply to high (low) MTB, using the acquirer market to book ratio. Sample medians are recomputed each year. The Fisher test of % *Variable Compensation* net effect is statistically significant for the low return on assets acquirer (column (2), Fisher statistics: 6.55, $p = 0.01$) and the high market to book (column (3), Fisher statistics: 2.87, $p = 0.09$) acquirer subsamples. The dependence between the % *Variable Compensation* net effect statistical significance and acquirer samples split according performance and valuation ratios is an additional indication that CEO incentives matter in the choice of the mode of payment. But in the absence of the detailed specification of the CEO compensation package (in particular, the financial indicators on which it depends), we have to remain cautious about the interpretation of coefficient signs obtained from subsamples.

6.2. Classic Determinants of M&A Mode of Payment

Can the marginalization of full stock payments in the U.S. be explained by the determinants of the M&A mode of payment classically used in M&A literature? To investigate this question, we study the predictive power of the determinants during the post-period,

Table 7. Determinants of the Probability of Full Stock Payment

	1990–2001		2002–2014		Diff. coeff.
	Coeff.	p-value	Coeff.	p-value	p-value
Panel A: United States—Pooling and post-pooling abolishment periods					
<i>Acquirer Size</i>	0.0263	(0.00)	−0.0484	(0.00)	(0.00)
<i>Relative Size</i>	0.0083	(0.42)	0.0366	(0.08)	(0.22)
<i>Acquirer Leverage</i>	−0.3784	(0.00)	0.0223	(0.67)	(0.00)
<i>Acquirer Market to Book</i>	0.0103	(0.00)	0.0056	(0.37)	(0.49)
<i>Acquirer Tangibility</i>	−0.1046	(0.00)	0.0427	(0.19)	(0.00)
<i>Acquirer Dividend</i>	−0.1115	(0.00)	−0.0406	(0.02)	(0.01)
<i>Acquirer Research and Development</i>	0.4103	(0.00)	0.4722	(0.00)	(0.95)
<i>Acquirer Cash</i>	−0.0090	(0.88)	−0.0040	(0.94)	(0.59)
<i>Domestic</i>	0.1128	(0.00)	0.0342	(0.06)	(0.01)
<i>Horizontal</i>	−0.0167	(0.35)	−0.0023	(0.88)	(0.55)
<i>Public Target</i>	0.0170	(0.36)	0.1471	(0.00)	(0.00)
<i>10-Year Interest Rate</i>	1.5629	(0.22)	3.8250	(0.00)	(0.15)
Sector fixed effects	Yes		Yes		
R ² (%)	15.93		19.10		
No. of observations	3,231		2,106		
Panel B: Canada—Pooling and post-pooling abolishment periods					
<i>Acquirer Size</i>	−0.0077	(0.83)	−0.0359	(0.00)	(0.41)
<i>Relative Size</i>	0.0147	(0.66)	−0.0039	(0.82)	(0.60)
<i>Acquirer Leverage</i>	−0.1382	(0.73)	−0.3044	(0.03)	(0.67)
<i>Acquirer Market to Book</i>	−0.0056	(0.88)	0.0239	(0.00)	(0.40)
<i>Acquirer Tangibility</i>	0.0063	(0.97)	0.0405	(0.47)	(0.83)
<i>Acquirer Dividend</i>	−0.2132	(0.11)	−0.1253	(0.01)	(0.51)
<i>Acquirer Research and Development</i>	2.3846	(0.02)	0.6039	(0.01)	(0.31)
<i>Acquirer Cash</i>	0.1973	(0.60)	−0.1634	(0.12)	(0.05)
<i>Domestic</i>	0.0226	(0.86)	0.1589	(0.00)	(0.26)
<i>Horizontal</i>	0.1804	(0.11)	0.0220	(0.55)	(0.15)
<i>Public Target</i>	0.1155	(0.28)	0.2975	(0.00)	(0.09)
<i>10-Year Interest Rate</i>	1.4213	(0.70)	0.7102	(0.68)	(0.85)
Sector fixed effects	Yes		Yes		
R ² (%)	26.65		35.70		
No. of observations	165		621		

Notes. This table displays the results of a linear probability model of full stock payment using the set of mode of payment choice determinants classically used in prior literature (Eckbo et al. 2017) by subperiod (pooling and post-pooling abolishment periods). Panel A reports the U.S. results, and panel B reports the results for Canada. All variables are defined in Table A.1. “Diff. coeff.” is a test of coefficient differences between the pooling and post-pooling abolishment periods, obtained by estimating a pooled specification with interaction terms.

using an empirical strategy similar to the one in Boone et al. (2014): we model the probability of full stock payments using an LPM specification and the set of determinants in Table 3. The estimates of the LPM model coefficients rely on the subsample of transactions that took place during the pooling period. We then use those estimated coefficients to obtain the fitted probabilities of full stock payments, during both the pooling and post-pooling periods. Finally, we average the deal

level fitted probabilities, year by year.²³ Table 7 displays the results, with the U.S. estimations in panel A and the Canadian estimations in panel B. In each panel, the left-hand column contains the estimation results for the pooling period, as used in this analysis. The average fitted probabilities are depicted in Figure 3, in addition to the observed frequencies of full stock payment. As is clearly apparent in Figure 3, panel A, the LPM model captures the average probability of full stock payment during the pooling period (in-sample) reasonably well for the United States, but it cannot explain the sharp decline during the post-pooling period. The comparison with the Canadian results (Figure 3, panel B) is striking. With the exception of the first few years (1990–1992), for which our sample is very limited (Section 2), the LPM captures the average probability of full stock payments correctly, but in this case it can do so for both the pooling and post-pooling periods. The analysis of Table 7, panel B, estimation results indicates that this is the case despite there being only one statistically significant independent variable in the Canadian case (essentially Acquirer Research and Development, with $p = 0.02$). These observations lead us to conclude that the known determinants of stock payments do not drive the full-stock-payment marginalization in the United States after 2001, in confirmation of the results of Boone et al. (2014). Because participation in the M&A market is a voluntary decision, endogenous self-selection may affect these results. We, therefore, replicated the exercise using a two-stage Heckman procedure, but the results were similar (available on request).

6.3. Classic Determinants of M&A Mode of Payment: Pooling vs. Post-Pooling Periods

The results in Table 7 also offer an opportunity to compare the classic determinants' coefficients across the pooling and post-pooling periods. In each panel of Table 7, we report estimates for the pooling (left) and post-pooling (right) periods, as well as the test of coefficient differences. In the United States (Table 7, panel A), the acquirer size, leverage, tangibility, and dividend coefficients change signs between periods, and the changes are statistically significant. The domestic and public target dummies keep their signs, but the changes in their coefficient values are significant. The pooling abolishment changed the profile of acquirers active in the M&A market, as well as the type of transactions undertaken. This result corroborates the prediction that a change in accounting regulations has material impacts on the M&A market. The situation in Canada is different. Results must be interpreted with caution because most explanatory variables are not statistically significant at the usual confidence levels. We observe, nevertheless, that significant evolutions appear for only two variable coefficients: acquirer cash

(which becomes negative and significant in the post-period) and the public target dummy (which is positive but only significant during the post-period).

6.4. Probability of Full Stock Payment and Pooling

If pooling was a main motivation to pay in stock during 1990–2001 in the United States, acquirers who would have chosen pooling should not pay more frequently in stock than other acquirers do after the pooling abolishment. We test this prediction using a two-stage approach. In the first stage, we estimate the probability of pooling during the pooling period. In the second stage, using fitted probabilities of pooling obtained from the first-stage estimates, we study whether the probability of pooling still explains full stock payments during the post-pooling period. The first-stage estimated equation is

$$Pooling_i = \Phi(\alpha + \varphi'(\mathbf{Controls}_i)), \quad (4)$$

where $Pooling_i$ is a dummy variable equal to 1 if the transaction was recognized under pooling, $\Phi(\cdot)$ indicates a probit specification, and $\mathbf{Controls}_i$ is a set of control variables that explains the choice to structure an acquisition to make it eligible for pooling. Ayers et al. (2002) highlight the role of the target's Step-Up, ROA, and leverage. This time, we select a probit specification, because there is no interaction term in Equation (4).

In the second stage, similar to the analyses reported in Table 5, we test two LPM specifications, one without year fixed effects and one with them:

$$\begin{aligned} Stock_i = & \alpha + \beta Post_{pool_i} + \gamma Pooling Hat_i \\ & + \delta(Post_{pool_i} \times Pooling Hat_i) + \theta'(\mathbf{Sector FE}_i) \\ & + \varphi'(\mathbf{Controls}_i) + \varepsilon_i, \quad \text{and} \end{aligned} \quad (5)$$

$$\begin{aligned} Stock_i = & \alpha + \gamma Pooling Hat_i + \delta(Post_{pool_i} \times Pooling Hat_i) \\ & + \gamma(\mathbf{Year FE}_i) + \theta'(\mathbf{Sector FE}_i) \\ & + \varphi'(\mathbf{Controls}_i) + \varepsilon_i. \end{aligned} \quad (6)$$

The variable $Pooling Hat$ refers to the fitted probabilities of pooling, obtained using Equation (4). Finally, δ , the coefficient of the interaction terms $Post_{pool_i} \times Pooling Hat_i$, is the coefficient of interest.

Table 8, panel A displays the results of the estimation of Equation (4), obtained using M&A transactions from 1990 to 2001 in the United States.²⁴ As suggested by Ayers et al. (2002), the target's Step-Up, ROA, and leverage have significant roles. Other control variables are also significant, including the target's size (positive coefficient) and the acquirer's dividend (negative coefficient), run up (positive coefficient), and manufacturing industries (negative coefficient).

Table 8, panel B contains the results of the estimations of Equations (5) and (6), which we obtained

Table 8. Probabilities of Full Stock Payments and of Pooling: U.S. Case

Panel A: First-stage probability of pooling				
	Coeff.		p-value	
Step Up	0.0215		(0.05)	
Target Return on Assets	0.8622		(0.01)	
Target Leverage	−0.9644		(0.00)	
Target Size	0.1653		(0.00)	
Relative Size	−0.0478		(0.48)	
Acquirer Size	0.0346		(0.50)	
Acquirer Leverage	−0.3756		(0.27)	
Acquirer Cash	0.2968		(0.51)	
Acquirer Market to Book	0.0473		(0.27)	
Acquirer Tangibility	−0.0412		(0.80)	
Acquirer Dividend	−0.2254		(0.08)	
Run Up	0.2123		(0.03)	
High Tech	−0.0333		(0.80)	
Manufacturing	−0.6501		(0.00)	
Pseudo R ² (%)		13.22		
No. of observations		729		
Panel B: Second-stage probability of full stock payment, linear probability model				
	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
$Post_{pool}$	0.1922	(0.01)		
Pooling Hat	0.9336	(0.00)	0.8419	(0.00)
$Post_{pool} \times Pooling\ Hat$	−1.1326	(0.00)	−0.9740	(0.00)
Acquirer Size	−0.0688	(0.00)	−0.0640	(0.00)
Relative Size	−0.0474	(0.00)	−0.0490	(0.00)
Acquirer Leverage	−0.1396	(0.15)	−0.1154	(0.23)
Acquirer Market to Book	0.0185	(0.10)	0.0200	(0.07)
Acquirer Tangibility	−0.0312	(0.56)	−0.0244	(0.66)
Acquirer Dividend	−0.0538	(0.13)	−0.0530	(0.14)
Acquirer Research and Development	0.3939	(0.01)	0.4337	(0.01)
Acquirer Cash	0.0791	(0.48)	0.1065	(0.33)
Domestic	0.1091	(0.14)	0.1456	(0.05)
Horizontal	−0.0465	(0.10)	−0.0433	(0.12)
Public Target	0.0023	(0.98)	0.0247	(0.85)
10-Year Interest Rate	1.5632	(0.38)	−7.2418	(0.14)
Credit Spread	1.1686	(0.26)	9.7964	(0.01)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
R ² (%)	27.85		29.66	
No. of observations	1,157		1,157	

Notes. This table reports the results of a two-stage analysis of the relation between the probability of full stock payment and the probability of pooling. The M&A sample for the United States is introduced in Table 1 but restricted to public targets; the variables are defined in Table A.1. Panel A presents the first-stage probability of the pooling analysis. The dependent variable is the pooling dummy variable (equal to 1 in the case of pooling). We adopt a classic probit specification. Control variables are selected in accordance with existing literature about pooling accounting (Ayers et al. 2002). The estimation is performed on the pooling period (1990–2001). Panel B is dedicated to the second-stage probability of the full-stock-payment analysis. The dependent variable is the full-stock-payment dummy variable. Both specifications rely on a linear probability model. In column (1) the $Post_{pool}$ dummy variable is explicitly introduced. In column (2) we introduce year fixed effects. $Pooling\ Hat$ is the fitted probability of pooling, obtained using the first-stage probability of the pooling analysis. $Post_{pool} \times Pooling\ Hat$ is the interaction term between the $Post_{pool}$ dummy variable and $Pooling\ Hat$. The set of control variables is the same as in Table 4. Standard errors are robust to heteroskedasticity.

using the sample of U.S. M&A transactions for which we could access all necessary information. The two key coefficients are those for $Pooling\ Hat_i$ and the interaction term, $Post_{pool_i} \times Pooling\ Hat_i$. The former is positive and highly significant in both specifications (column (1), 0.9336, $p = 0.00$; column (2), 0.8419, $p = 0.00$). Acquirers interested in pooling selected full stock payments more often during the pooling period. This finding must be the case, because full stock payment was a necessary condition for pooling (Section 4). The coefficient of $Post_{pool_i} \times Pooling\ Hat_i$ is negative and also highly significant (column (1), −1.1326, $p = 0.00$; column (2), −0.9740, $p = 0.00$). When we add the coefficients of $Pooling\ Hat_i$ and $Post_{pool_i} \times Pooling\ Hat_i$ to obtain the net effect of $Pooling\ Hat_i$ during the post-pooling period, we obtain a value close to 0 (not significantly different from 0; unreported result). This is the anticipated result if pooling was a main reason to select full stock payment. That is, acquirers wishing to opt for pooling had no more opportunity to do so during the post-pooling period, so they stopped paying in stock more frequently than other acquirers.

6.5. Value Effects

Fuller et al. (2002) and Officer et al. (2009) show that the mode of payment interacts with the target status to determine acquirer value creation in M&A transactions. The most value-creating transactions for acquirers are those of private targets paid in stock. Private target acquisitions are more value-creating for acquirers than are public ones, possibly because of the presence of an illiquidity premium captured by acquirers. Moreover, relative to cash payments, the use of stock payment enables acquirers to transfer part of target valuation uncertainty to target shareholders (more so for private targets because of the lack of a market price). The acquisition of public targets paid in stock instead is the most value-destroying combination, because it combines a size effect with a share exchange offer, both of which generate negative investor reactions (Golubov et al. 2016, Moeller et al. 2004).

Our results show that the abolishment of pooling in 2001 (SFAS 141) represents a main explanation of full-stock-payment marginalization in the United States after 2001. In Section 6.3 we detailed how the stock-paid transaction characteristics and corresponding acquirer profiles changed after the pooling abolishment. Here, we investigate whether these changes affected investors' perceptions of the value creation surrounding the transactions.

We start by computing the acquirer CAR for our U.S. sample (see Section 2). We use the standard market model as a return-generating process, estimated on a window from day −300 to day −90, with respect to the announcement date. The acquirer CAR are for a three-day event window, centered on the announcement

date. We then regressed acquirer CAR on the set of classical determinants (Golubov et al. 2015) and dummy variables that capture the post-pooling period ($Post_{pool}$), the target's status (*Public Target*), and full stock payment (*Stock*).

Table 9 contains the results. Column (1) includes only the classic acquirer CAR determinants for comparison with existing results. Similar to Fuller et al. (2002) and Officer et al. (2009), we observe that acquiring public targets is negatively perceived by investors, and the effect is reinforced by full stock payments. We add the dummy variables— $Post_{pool}$, *Public Target*, and *Stock*—and their interactions in column (2). The $Post_{pool} \times Stock \times Public\ Target$ interaction coefficient is negative and highly significant (-0.0463 , $p = 0.01$), indicating that investors react far more negatively to acquisition announcements for public targets with full payment in stock after pooling abolishment. Noting that fully stock-paid transactions display significant changes in acquirer profiles and deal characteristics

in the post-pooling period (Section 6.3), we determine that the pooling abolishment has transformed this segment of M&A activity. Only less value-creating or more value-destroying transactions of public targets continue to be fully paid in stock, and full stock payment conveys more negative private information about acquirers.

7. Robustness Checks

We investigate the robustness of the Table 5 results, reflecting our baseline difference-in-differences test of the pooling abolishment hypothesis, to various issues that could lead to misleading conclusions. To start, 2001 was the year SFAS 141 and 142 were adopted, but also the Internet bubble burst year. Did this contemporaneous event drive our results? In Table 10, panel A we present a replication of the Table 5 tests after exclusion of high-tech firms, which we identify using Kile and Phillips's (2009) method. These authors provide a detailed list of four-digit SIC industries that can be considered high-tech. After excluding them, our sample size dropped from 6,123 observations in Table 5 to 3,273 observations in Table 10. The coefficients of the $US \times Post_{pool}$ dummy interaction term remained negative and highly significant (column (1), -0.2410 , $p = 0.00$; column (2), -0.1777 , $p = 0.00$). The bursting of the Internet bubble thus is not a confounding factor. This result also indicates that pooling relevance during the 1990s was not restricted to high-tech industries, as was frequently suggested by the financial press at this time.

If cross-border transactions are asymmetrically distributed between the United States and Canada and also underwent specific time trends after 2001, they also could act as confounding factors. We replicated our preceding analysis, but this time we excluded cross-border transactions. The results in Table 10, panel B, stem from a sample size of 5,174 transactions. The coefficients of the $US \times Post_{pool}$ dummy interaction term became even more negative and highly significant (column (1), -0.3693 , $p = 0.00$; column (2), -0.2214 , $p = 0.00$).

The pooling and post-pooling subsamples of acquirers each incorporate a reduced number of identical firms. As we reported in Section 6.3, in the United States, the determinants of full stock payment underwent significant changes between the two periods. Might this variation in the composition of the acquirer samples affect the estimation results? To check, we selected a subsample of U.S. acquirers that undertook at least one transaction in the five years before the pooling abolishment and another transaction in the five years following it. We refer to this subsample as the constant acquirer sample, and we present the pertinent results in Table 10, panel C. The sample size dropped drastically to 971 observations. The coefficients of the $US \times Post_{pool}$ dummy interaction term remain negative

Table 9. Method of Payment, Pooling Abolishment, and Value Effects: U.S. Case

	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
$Post_{pool}$			-0.0033	(0.50)
$Post_{pool} \times Stock$			0.0116	(0.39)
$Post_{pool} \times Public\ Target$			0.0132	(0.07)
$Post_{pool} \times Stock \times Public\ Target$			-0.0463	(0.01)
<i>Stock</i>	0.0087	(0.10)	0.0081	(0.18)
<i>Public Target</i>	-0.0190	(0.00)	-0.0272	(0.00)
<i>Stock</i> $\times$ <i>Public Target</i>	-0.0365	(0.00)	-0.0242	(0.01)
<i>Relative Size</i>	0.0187	(0.00)	0.0192	(0.00)
<i>Acquirer Size</i>	-0.0022	(0.18)	-0.0033	(0.02)
<i>Acquirer Market to Book</i>	-0.0002	(0.75)	-0.0005	(0.48)
<i>Acquirer Free Cash Flow</i>	-0.0016	(0.90)	-0.0013	(0.92)
<i>Domestic</i>	-0.0007	(0.87)	-0.0004	(0.94)
<i>Horizontal</i>	-0.0025	(0.47)	-0.0023	(0.52)
<i>Sigma</i>	0.5220	(0.02)	0.3543	(0.08)
<i>Run Up</i>	-0.0013	(0.68)	-0.0007	(0.82)
<i>Hostile</i>	-0.0033	(0.83)	-0.0046	(0.78)
<i>Tender Offer</i>	0.0123	(0.02)	0.0135	(0.01)
Sector fixed effects	Yes		Yes	
Year fixed effects	Yes		No	
R^2 (%)	8.58		7.63	
No. of observations	5,148		5,148	

Notes. This table reports the results of cross-sectional regressions of acquirer CAR on a large set of determinants classically used in prior literature (Golubov et al. 2015), dummy variables identifying the post-pooling abolishment period ($Post_{pool}$), fully stock paid acquisitions (*Stock*), public targets (*Public Target*), and their interactions. The M&A sample for the United States is introduced in Table 1, and the variables are defined in Table A.1. The CAR are obtained using the market model as a return-generating process (estimation window from day -300 to day -90 relative to the announcement date) and a three-day event window centered on the announcement date. Column (1) displays the results for the baseline specification. Column (2) adds the dummy variables of interest and their interactions. Standard errors are robust to heteroskedascity.

Table 10. Robustness Checks

Panel A: Internet bubble				
	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
<i>US</i>	0.1484	(0.00)	0.1441	(0.00)
<i>Post_{pool}</i>	0.0426	(0.41)		
<i>US × Post_{pool}</i>	−0.2410	(0.00)	−0.1777	(0.00)
<i>Acquirer Size</i>	−0.0083	(0.30)	−0.0074	(0.36)
<i>Relative Size</i>	−0.0120	(0.03)	−0.0092	(0.10)
<i>Acquirer Leverage</i>	−0.2888	(0.00)	−0.2877	(0.00)
<i>Acquirer Market to Book</i>	0.0237	(0.01)	0.0247	(0.00)
<i>Acquirer Tangibility</i>	−0.0246	(0.37)	−0.0289	(0.29)
<i>Acquirer Dividend</i>	−0.0684	(0.00)	−0.0702	(0.00)
<i>Acquirer Research and Development</i>	0.8465	(0.00)	0.8396	(0.00)
<i>Acquirer Cash</i>	−0.0673	(0.31)	−0.0746	(0.26)
<i>Domestic</i>	0.1236	(0.00)	0.1202	(0.00)
<i>Horizontal</i>	−0.0107	(0.52)	−0.0129	(0.44)
<i>Public Target</i>	0.1021	(0.00)	0.1017	(0.00)
<i>10-Year Interest Rate</i>	1.7386	(0.05)	−1.8921	(0.35)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
<i>R</i> ² (%)	20.46		21.16	
No. of observations	3,273		3,273	
Panel B: Domestic deals				
	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
<i>US</i>	0.1937	(0.00)	0.1291	(0.00)
<i>Post_{pool}</i>	0.0804	(0.13)		
<i>US × Post_{pool}</i>	−0.3693	(0.00)	−0.2214	(0.00)
<i>Acquirer Size</i>	0.0077	(0.38)	0.0091	(0.30)
<i>Relative Size</i>	−0.0023	(0.59)	−0.0009	(0.83)
<i>Acquirer Leverage</i>	−0.2232	(0.00)	−0.2203	(0.00)
<i>Acquirer Market to Book</i>	0.0167	(0.00)	0.0189	(0.00)
<i>Acquirer Tangibility</i>	−0.0366	(0.12)	−0.0392	(0.10)
<i>Acquirer Dividend</i>	−0.0838	(0.00)	−0.0825	(0.00)
<i>Acquirer Research and Development</i>	0.4474	(0.00)	0.4346	(0.00)
<i>Acquirer Cash</i>	−0.0058	(0.89)	−0.0071	(0.87)
<i>Horizontal</i>	−0.0054	(0.68)	−0.0075	(0.57)
<i>Public Target</i>	0.0784	(0.00)	0.0766	(0.00)
<i>10-Year Interest Rate</i>	2.8727	(0.00)	−0.6842	(0.70)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
<i>R</i> ² (%)	23.19		23.96	
No. of observations	5,174		5,174	
Panel C: Constant acquirer sample, between [−5, 0] and [0, +5] year ranges				
	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
<i>US</i>	0.2464	(0.00)	0.1412	(0.00)
<i>Post_{pool}</i>	0.0090	(0.91)		
<i>US × Post_{pool}</i>	−0.3431	(0.00)	−0.2079	(0.00)
<i>Acquirer Size</i>	−0.0009	(0.96)	−0.0035	(0.84)
<i>Relative Size</i>	−0.0140	(0.23)	−0.0147	(0.21)
<i>Acquirer Leverage</i>	−0.3213	(0.00)	−0.3137	(0.01)
<i>Acquirer Market to Book</i>	0.0103	(0.00)	0.0139	(0.00)
<i>Acquirer Tangibility</i>	−0.1070	(0.11)	−0.1040	(0.12)
<i>Acquirer Dividend</i>	−0.0302	(0.47)	−0.0319	(0.43)
<i>Acquirer Research and Development</i>	0.6285	(0.00)	0.5594	(0.00)
<i>Acquirer Cash</i>	−0.1349	(0.20)	−0.1155	(0.25)
<i>Domestic</i>	0.0994	(0.01)	0.1051	(0.01)
<i>Horizontal</i>	−0.0085	(0.78)	−0.0111	(0.72)
<i>Public Target</i>	0.0645	(0.04)	0.0649	(0.04)
<i>10-Year Interest Rate</i>	3.2220	(0.23)	4.5415	(0.36)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
<i>R</i> ² (%)	27.83		30.22	
<i>N</i>	971		971	

Table 10. (Continued)

Panel D: Controlling for cross-listed Canadian firms				
	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
US	0.1879	(0.00)	0.1442	(0.00)
Post _{pool}	0.0283	(0.53)		
US × Post _{pool}	−0.3231	(0.00)	−0.2147	(0.00)
Canadian Cross-Listed	0.0707	(0.05)	0.0625	(0.08)
Acquirer Size	0.0046	(0.53)	0.0047	(0.53)
Relative Size	−0.0064	(0.11)	−0.0046	(0.24)
Acquirer Leverage	−0.2186	(0.00)	−0.2166	(0.00)
Acquirer Market to Book	0.0175	(0.00)	0.0193	(0.00)
Acquirer Tangibility	−0.0480	(0.03)	−0.0494	(0.02)
Acquirer Dividend	−0.0815	(0.00)	−0.0817	(0.00)
Acquirer Research and Development	0.4479	(0.00)	0.4345	(0.00)
Acquirer Cash	−0.0164	(0.68)	−0.0153	(0.70)
Domestic	0.1035	(0.00)	0.1013	(0.00)
Horizontal	−0.0046	(0.70)	−0.0066	(0.58)
Public Target	0.0846	(0.00)	0.0814	(0.00)
10-Year Interest Rate	2.1051	(0.00)	−1.2655	(0.42)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
R ² (%)	23.51		24.23	
No. of observations	6,123		6,123	

Panel E: European countries as counterfactual								
	(1)		(2)		(3)		(4)	
	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value
US	0.3001	(0.00)	0.2986	(0.00)	0.2782	(0.00)	0.2781	(0.00)
Post _{pool}	−0.0679	(0.00)			−0.0560	(0.00)		
US × Post _{pool}	−0.2221	(0.00)	−0.2241	(0.00)	−0.2214	(0.00)	−0.2238	(0.00)
Deal Size					−0.0065	(0.01)	−0.0063	(0.01)
Domestic					0.0739	(0.00)	0.0724	(0.00)
Horizontal					−0.0081	(0.33)	−0.0074	(0.37)
Public Target					0.0765	(0.00)	0.0729	(0.00)
Sector fixed effects	Yes		Yes		Yes		Yes	
Year fixed effects	No		Yes		No		Yes	
R ² (%)	14.12		14.87		15.03		15.70	
No. of observations	13,235		13,235		13,235		13,235	

Panel F: United Kingdom as counterfactual								
	(1)		(2)		(3)		(4)	
	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value
US	0.3472	(0.00)	0.3402	(0.00)	0.3419	(0.00)	0.3359	(0.00)
Post _{pool}	−0.0653	(0.00)			−0.0534	(0.00)		
US × Post _{pool}	−0.2278	(0.00)	−0.2166	(0.00)	−0.2258	(0.00)	−0.2150	(0.00)
Deal Size					−0.0167	(0.00)	−0.0161	(0.00)
Domestic					0.0538	(0.00)	0.0503	(0.00)
Horizontal					−0.0190	(0.04)	−0.0187	(0.04)
Public Target					0.0827	(0.00)	0.0791	(0.00)
Sector fixed effects	Yes		Yes		Yes		Yes	
Year fixed effects	No		Yes		No		Yes	
R ² (%)	16.78		17.41		17.63		18.17	
No. of observations	10,629		10,629		10,629		10,629	

Panel G: Treatment effect test with propensity score matching				
Treatment effect test	No. of observations		Mean	p-value
Average treatment effect				
Post _{pool} (1 vs. 0)	3,817		−0.3320	(0.00)

Table 10. (Continued)

Panel H: Difference-in-differences test, controlling for SEC Rule 10b-18				
	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
<i>US</i>	0.1652	(0.00)	0.1258	(0.00)
<i>Post_{pool}</i>	0.0257	(0.70)		
<i>Post_{purch}</i>	−0.0405	(0.50)		
<i>US × Post_{pool}</i>	−0.2752	(0.00)	−0.1673	(0.00)
<i>US × Post_{purch}</i>	−0.0403	(0.52)	−0.0727	(0.09)
<i>Acquirer Size</i>	0.0048	(0.52)	0.0050	(0.50)
<i>Relative Size</i>	−0.0050	(0.20)	−0.0036	(0.37)
<i>Acquirer Leverage</i>	−0.2189	(0.00)	−0.2154	(0.00)
<i>Acquirer Market to Book</i>	0.0173	(0.00)	0.0190	(0.00)
<i>Acquirer Tangibility</i>	−0.0467	(0.03)	−0.0467	(0.03)
<i>Acquirer Dividend</i>	−0.0811	(0.00)	−0.0829	(0.00)
<i>Acquirer Research and Development</i>	0.4449	(0.00)	0.4352	(0.00)
<i>Acquirer Cash</i>	−0.0139	(0.72)	−0.0143	(0.71)
<i>Domestic</i>	0.1008	(0.00)	0.0978	(0.00)
<i>Horizontal</i>	−0.0046	(0.70)	−0.0065	(0.58)
<i>Public Target</i>	0.0820	(0.00)	0.0816	(0.00)
<i>10-Year Interest Rate</i>	1.2432	(0.11)	−0.5847	(0.72)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
<i>R</i> ² (%)	23.59		24.23	
No. of observations	6,123		6,123	

Notes. This table summarizes the robustness checks of the results in Table 5, our baseline difference-in-differences test of the pooling abolishment hypothesis. In panel A we excluded high-technology firms (Kile and Phillips 2009). In panel B we restrict the sample to domestic deals. In panel C we restrict the sample to U.S. acquirers that undertook at least one transaction in the five years before and another transaction in the five years after pooling abolishment. In panel D we control for Canadian cross-listed firms. In panel E (panel F) we report the results of our difference-in-differences test specifications using Europe (the United Kingdom) as a counterfactual. In panel G we report the results of a treatment effect test with propensity score matching to avoid the use of non-U.S. transactions as counterfactual. In panel H we report an additional difference-in-differences test controlling for the 2003 SEC amendments to the share repurchases safe harbor.

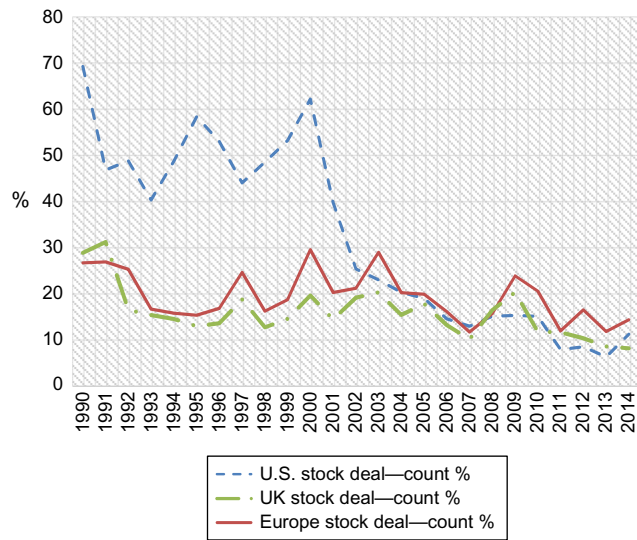
and highly significant again (column (1), -0.3431 , $p = 0.00$; column (2), -0.2079 , $p = 0.00$).

Many Canadian companies are listed on U.S. capital markets, as well as Canadian markets, which defines them as cross-listed firms. If the cross-listing decision was motivated by the search for easier conditions for pooling in the United States, the presence of these cross-listed firms could affect our results.²⁵ We check the robustness of our Table 5 results to this issue by adding a dummy variable, *Canadian Cross-Listed*, equal to 1 if a firm was cross-listed. According to Table 10, panel D, the coefficients of the $US \times Post_{pool}$ dummy interaction effect remains negative and highly significant (column (1), -0.3231 , $p = 0.00$; column (2), -0.2147 , $p = 0.00$). An alternative robustness test dropped Canadian cross-listed firms from the sample; in doing so, we obtained comparable results (available on request) but a drastically reduced Canadian firm sample size.

The use of pooling was very limited in European countries, whether because it was forbidden by country-specific regulations or because of their restrictive conditions, similar to those in Canada (Amel-Zadeh et al. 2015).²⁶ Europe, therefore, qualifies as another

valid counterfactual for our difference-in-differences test. We extracted, from Thomson SDC, M&A transactions completed by European acquirers, using the same criteria that we describe in Section 2. The resulting sample of 6,285 transactions involves companies in 41 European countries, though UK firms account for more than 58% of them. In Figure 4, we present the yearly average percentages of fully stock-paid transactions, in count-based values, for European and UK samples compared with the U.S. one. The European and UK patterns are very different from the U.S. one, displaying no systematic decline after 2001. For example, the yearly average stock-paid transaction percentages for Europe (UK) were 20.76% (17.44%) during 1990–2001 and then 17.59% (13.77%) during 2002–2014. The parallel trends assumption underlying a valid difference-in-differences test implementation is clearly respected again; the European and UK samples displayed behaviors similar to the U.S. sample both before and after the pooling abolishment. The U.S. post-pooling abolishment fully stock-paid transaction frequency also rejoined the long-term averages for Europe; this observation should prompt further investigation. Table 10,

Figure 4. (Color online) Percentage of Stock-Paid M&A Transactions in the United States, Europe, and the United Kingdom During 1990–2014



Notes. This figure displays the percentage of fully stock-paid M&A transactions in the United States, Europe, and the United Kingdom from 1990 to 2014 (count-based). The sample is collected from the Thomson SDC database, using the selection criteria listed in Section 2, which lead to sample sizes of 6,955 U.S. transactions and 6,285 European (41 countries) transactions, of which 3,679 are UK transactions.

panel E provides the results for Europe (including the United Kingdom); panel F focuses on the subsample of UK acquirers. In panel E, column (1), the coefficient value is -0.2221 ($p = 0.00$), and in column (2), it is -0.2241 ($p = 0.00$). These estimates strongly support the pooling abolishment hypothesis, because in contrast with the European experience, full stock payments declined significantly in the United States after the pooling abolishment. The result is also robust to the introduction of the control variables,²⁷ industry-level latent factors constant through time, and annual common latent factors.²⁸

A main contribution of our analyses is the demonstrated use of difference-in-differences tests with valid counterfactuals. In Section 4, we detailed the U.S. and Canadian accounting regulations to explain why Canada was so well suited to serve as a counterfactual; in this section, we describe how Europe is qualified as well. We finally apply a classic treatment effect test for the pooling abolishment hypothesis, avoiding non-U.S. transactions as the counterfactual. Specifically,

—for a given M&A transaction, we define the treatment as the announcement date taking place during the post-period;

—we use propensity score-matching estimators to impute the missing counterfactual for each transaction as the outcome of the most similar transaction that also took place during the pooling period. The outcome of this transaction is the potential outcome;

—we compute the treatment effect by taking the average of the differences between the observed and potential outcomes for each subject.

A detailed presentation of this procedure is available from Roberts and Whited (2013). We used acquirer and transaction characteristics to estimate the treatment model. Table 10, panel G displays the treatment effect test results: the average treatment effect is negative and strongly significant (-0.3320 , $p = 0.00$), consistent with our results in Table 5. Let us also mention that we replicate this exercise using the same set of control variables as in Ali and Kravet (2016). Our sample is this time reduced to 1,159 observations because of limited data availability for unlisted targets. We obtain comparable results (available on request).

Rule 10b-18 under the Securities Exchange Act of 1934 creates a safe harbor from claims of market manipulation in the case of share repurchases (see the Simpson Thacher report²⁹ for limitations and restrictions that apply to Rule 10b-18). In November 2003, the SEC drastically restricted the applicability of this share repurchases safe harbor for most noncash M&A. The 2003 ruling did not prohibit share repurchases while M&A transactions were pending, but it significantly increased the risk of in-depth regulatory scrutiny in such cases. The November 2003 amendment applied only to noncash transactions, such that it created new incentives for acquirers to pay in cash and thereby benefit from Rule 10b-18's safe harbor. Accordingly, we test whether our results are robust to this regulatory change by including a $Post_{purch}$ dummy variable (equal to 1 for transactions announced after November 2003) and an additional $US \times Post_{purch}$ interaction term in our difference-in-differences specification. As the results in Table 10, panel H, reveal, the $US \times Post_{pool}$ coefficient remains negative and strongly significant in both difference-in-differences specifications, whereas the $US \times Post_{purch}$ interaction coefficient is significant only in the specification with year fixed effects ($p = 0.09$). The previous results are thus robust to the SEC Rule 10b-18 share repurchases safe harbor amendments of 2003.

8. Conclusion

This analysis of the evolution of full stock payments in M&A transactions in the United States in recent decades shines a light on the striking, sharp decline in the percentage of transactions fully paid in stock after 2001. The consistency of about 10% of stock-paid transactions since 2010 suggests that we are not exaggerating when we note the marginalization of this payment mode. Because 2001 is the year that pooling and goodwill amortization were abolished in the United States (SFAS 141 and 142), we test whether this accounting rule change could have had such far-reaching implications. The question is particularly relevant, because

the choice of a mode of payment in M&As represents a sort of laboratory to test various theoretical predictions from corporate finance.

Testing for the effects of pooling abolishment is challenging because it happened just once. Therefore, convincing empirical evidence requires a carefully chosen identification strategy. We use a difference-in-differences test with Canada as a counterfactual for the U.S. experience. The Canadian economy is closely tied to the U.S. economy, and both share many common latent factors. Moreover, the evolution of Canada's pooling regulations creates a particularly interesting way to test the pooling hypothesis. Pooling was also abolished in 2001 in Canada, but it had rarely been used previously, unlike in the United States. Using Canada as a placebo for the pooling abolishment treatment in the United States, we test whether the U.S. abolishment was an effective medication (explain the decreasing in full stock payment). With large samples of M&A transactions between 1990 and 2014, our results clearly support the notion that pooling abolishment dramatically reduced full stock acquisitions. Our complementary analyses reveal that CEO incentives contribute to the impact of pooling abolishment.

Implications for Policy and Practice

Pooling accounting and goodwill amortization were abolished in 2001 by FASB 141 and 142. The motivations mentioned in FASB exposure drafts were (a) to remove the

burden of assuring compliance with the 12 criteria previously required for pooling accounting, (b) to promote a uniform treatment of all M&A transactions, (c) to comply more closely with international standards, (d) to improve information quality and comparability, and (e) to foster better managerial incentives. Our paper finds that the FASB rulings had a major impact on the mode of payment in M&A transactions. After pooling was abolished, payment fully in stock decreased from 62% of the transactions in 2000 to 25% in 2002. Evidently, this was an unanticipated consequence of FASB 141 and 142. This experience reveals the importance of regulations for corporate decision makers, even when such regulations have no obvious impact on cash flows.

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Appendix

Table A.1. Variable Definitions

Variable	Definition	Source
% Variable Compensation	Variable component of the acquirer CEO's compensation: (item TDC1-item SALARY)/item TDC1	Execucomp
10-Year Interest Rate	10-year government bond (United States and Canada) interest rate	FRED
Acquirer CAR	Acquirer CAR over the three-day event windows centered on the announcement date, estimated with a market model and with an estimation window from day -300 to day -90; CRSP value weighted index is used as proxy for the market index	crsp, sdc
Acquirer Cash	Cash on total asset (item CH/item AT)	Compustat
Acquirer Dividend	Dummy equal to 1 if bidder paid dividend previous year and 0 otherwise	Compustat
Acquirer Free Cash Flow	Income before extraordinary items (item IBC) divided by total assets (item AT)	Compustat
Acquirer Leverage	Acquirer long-term debt (item DLTT) divided by total assets (item AT)	Compustat
Market to Book	Total assets minus common equity (item CEQ) plus the market value of equity (item CSHO × item PRCC_F) divided by total assets (item AT)	Compustat
Acquirer Research and Development	In process Research and Development Expense (item RDIP) on total assets (item AT)	Compustat
Acquirer Return on Assets	Acquirer Earnings Before Interests and Taxes (item EBIT) on total assets (item AT)	Compustat
Acquirer Size	Market value of bidder 42 days before announcement (logarithm is used in multivariate analyses)	Compustat, SDC, DS
Acquirer Tangibility	Property, plant and equipment total (item PPEGT) on total asset (item AT)	Compustat
Canadian Cross-Listed	Dummy variable equal to 1 if the Canadian is also listed in the United States	Compustat
Credit Spread	Spread Moody corporate aaa	FRED
Deal Size	Deal value in millions USD	SDC
Domestic	Dummy equal to 1 if the acquirer and the target country are the same and 0 otherwise	SDC

Table A.1. (Continued)

Variable	Definition	Source
<i>High Tech</i>	Dummy variable equal to 1 if the acquirer is in the high-technology sector (Kile and Phillips 2009) and 0 otherwise	SDC
<i>Horizontal</i>	Dummy variable equal to 1 if the acquirer and the target belong to the same four-digit SIC code and 0 otherwise	SDC
<i>Hostile</i>	Dummy variable equal to 1 if the deal is classified as hostile by SDC and 0 otherwise	SDC
<i>Manufacturing</i>	Dummy variable equal to 1 if the acquirer is in manufacturing sector (SIC codes 2000–3999, high-technology sectors excluded)	SDC
<i>Pooling Hat</i>	Estimated probabilities of pooling (see Equation (6))	SDC
$Post_{pool}$	Dummy equal to 1 if announcement date of the deal is after 2001 (after June 30, 2001, for daily data) and 0 otherwise	SDC
$Post_{purch}$	Dummy equal to 1 if announcement date of the deal is after October 2003 and 0 otherwise	
<i>Public Target</i>	Dummy variable equal to 1 if the target is a public firm and 0 otherwise	SDC
<i>Relative Size</i>	Ratio of deal value on acquirer market value computed in day –42	SDC Compustat, CRSP Datastream CRSP
<i>Run Up</i>	Market-adjusted buy and hold return of the acquirer's stock over a 200-day window (–210, –11)	
<i>Sector fixed effects</i>	Sector fixed effects (two-digit SIC codes)	SDC
<i>Sigma</i>	Standard deviation of the market adjusted daily returns of the acquirer stock over a 200-day window (from day –210 to day –11 relative to the announcement date)	CRSP
<i>Step Up</i>	Step-up in target book value, equals to deal value minus target book equity (Compustat item CEQ) divided by target book equity	SDC, Compustat
<i>Stock</i>	Dummy variable equal to 1 if the consideration is stock only and 0 otherwise	SDC
<i>Target Leverage</i>	Target long-term debt (item DLTT) divided by total assets (item AT)	Compustat
<i>Target Return on Assets</i>	Target earnings before interests and taxes (item EBIT) on total assets (item AT)	Compustat
<i>Target Size</i>	Market value of target 42 days before announcement (logarithm is used in multivariate analyses)	SDC, CRSP, Datastream
<i>Tender Offer</i>	Dummy variable equal to 1 if the deal is classified as a tender offer by SDC and 0 otherwise	SDC
<i>Trend</i>	Linear time trend variable (increase of one unit each year)	
<i>US</i>	Dummy equal to 1 if acquirer country is the United States and 0 otherwise	SDC
<i>Year fixed effects</i>	Year fixed effects build on the year of deal announcement	SDC

Notes. SDC: Thomson SDC M&A database; CRSP: Center for Research in Security Prices database; Compustat: Compustat Fundamental Annual database and Compustat North America; DS: Datastream database; Execucomp: Compustat Execucomp database; FRED: Federal Reserve Economic Data.

Table A.2. Method of Payment and Pooling Abolishment, Probit-Based Results

	(1)		(2)	
	Coeff.	p-value	Coeff.	p-value
<i>US</i>	0.4350	(0.00)	0.3449	(0.00)
$Post_{pool}$	0.0940	(0.51)		
$US \times Post_{pool}$	–1.0183	(0.00)	–0.6918	(0.00)
<i>Acquirer Size</i>	0.0127	(0.55)	0.0141	(0.51)
<i>Relative Size</i>	–0.0272	(0.03)	–0.0210	(0.11)
<i>Acquirer Leverage</i>	–0.6620	(0.00)	–0.6595	(0.00)
<i>Acquirer Market to Book</i>	0.0627	(0.00)	0.0684	(0.00)
<i>Acquirer Tangibility</i>	–0.1210	(0.09)	–0.1316	(0.07)
<i>Acquirer Dividend</i>	–0.3022	(0.00)	–0.3057	(0.00)
<i>Acquirer Research and Development</i>	1.7142	(0.00)	1.6388	(0.00)
<i>Acquirer Cash</i>	–0.0712	(0.57)	–0.0583	(0.64)
<i>Domestic</i>	0.3525	(0.00)	0.3501	(0.00)
<i>Horizontal</i>	–0.0097	(0.80)	–0.0157	(0.69)
<i>Public target</i>	0.3161	(0.00)	0.3051	(0.00)
<i>10-Year Interest Rate</i>	8.5784	(0.00)	–4.4859	(0.37)
Sector fixed effects	Yes		Yes	
Year fixed effects	No		Yes	
Pseudo R^2 (%)	20.19		20.97	
No. of observations	6,100		6100	

Notes. This table reproduces Table 5, using a probit specification. The dependent variable is the full-stock-payment dummy variable (equal to 1 if the transaction is fully paid in stock). The M&A sample for the United States and Canada is introduced in Table 1, and the variables are defined in Table A.1. In column (1) the $Post_{pool}$ dummy variable (equal to 1 for the post-pooling abolishment period) is explicitly introduced. In column (2) we introduce year fixed effects. The set of control variables is the same as in Table 4.

Endnotes

¹ Amel-Zadeh et al. (2015), for example, referencing several prior contributions, state, inter alia, that “Four historical developments are at work. First, a large and increasing proportion of the purchase consideration in M&A consists not of cash but of acquirers’ shares.”

² More accurately, pooling was not really an option but rather the consequence of fulfilling 12 criteria (see Section 3). Acquirers could structure their transactions to meet these criteria though.

³ Purchase accounting in the case of fully stock-paid acquisition is most often depicted in the literature as penalizing financial ratios such as earnings per share (EPS), return on equity (ROE) or return on assets (ROA) as compared to pooling (see Reda 1999). We thank an anonymous referee for pointing out that, in fact, it depends on the valuation and operating performance ratios of the acquirer and the target.

⁴ For a classic case of spurious regression in time-series analysis, see Granger and Newbold (1974).

⁵ Canadian Institute of Chartered Accountants (continuously revised) CICA Handbook: I, Accounting; II, Auditing. Canadian Institute of Chartered Accountants, Toronto.

⁶ All databases were accessed through Wharton Research Data Services (WRDS): <https://wrds-web.wharton.upenn.edu>.

⁷ Sample sizes vary from analysis to analysis, depending on the control variables and data availability constraints.

⁸ See <https://research.stlouisfed.org/fred2/>.

⁹ We obtain similar results when using value-based fully stock-paid percentages.

¹⁰ We drop the years to 1990–1993 for the Canadian analysis to avoid the potential for bias if the results depended on years with only a limited number of recorded transactions.

¹¹ Note that valid interpretation of interaction terms in nonlinear models remains a subject of debate (Kolasinski and Siegel 2010).

¹² We cannot compare the results for the public target dummy, because Eckbo et al. (2017) also include an interaction term with the target premium.

¹³ Note that SFAS 141 evolved in 2008 to become SFAS 141R. The purchase method was relabeled the acquisition method, and the changes made the acquisition method less attractive in some circumstances. But according to Ali and Kravet (2016), this change affected only a small minority of transactions.

¹⁴ SFAS 142 replaced goodwill amortization with impairments, based on yearly assessments of goodwill value.

¹⁵ The remaining nine criteria imposed strict restrictions on the voting right changes for common stocks.

¹⁶ We thank an anonymous referee for providing us with a very clear analysis of this issue.

¹⁷ It must be noted, however, that the number of new shares issued depends on the acquirer’s market value and the bid premium. In times of high market valuations, the increase in number of shares outstanding may be less.

¹⁸ Regressing the percentage of fully stock-paid transactions reported in Ali and Kravet (2016, Table 1) on a post-pooling dummy variable (equal to 1 in 2002 and onward) reveals a coefficient not statistically different from 0.

¹⁹ According to Farrell and Beechy (2002, p. 92), “if one company can be identified as the acquirer, then there is a widespread agreement that the purchase method should be used.”

²⁰ These results are confirmed in Table A.2 in the appendix, using a probit specification. The interpretation of the interaction term coefficients must be handled with care in this nonlinear context (Greene 2010), but we note that the coefficients are negative and highly significant in both specifications (column (1), -1.0183 , $p = 0.00$; column (2), -0.6918 , $p = 0.00$).

²¹ To further investigate the potential confounding role of the pooling abolishment market discipline effect, we replicate the analysis adding the bid premium as an additional control variable. As this variable is mostly available only for the U.S. transactions, this robustness check relies on a treatment effect approach. We obtain qualitatively similar results as the ones reported in Table 5 (available on request). This evidence confirms that our results are robust to the potential market discipline effect.

²² We cannot replicate the Table 5 difference-in-differences specification, because CEO compensation packages are not available for Canadian acquirers.

²³ However, in contrast with Boone et al. (2014), our sample includes private target acquisitions, and fitted probabilities are strictly based on out-of-sample predictions for the post-pooling period.

²⁴ The significant sample size reduction is due to the need for data for computing target firm variables, such as the Step-Up and ROA.

²⁵ Canadian cross-listed firms did not use pooling more often than single-listed Canadian firms before the abolishment, an observation which is inconsistent with the idea that pooling might have motivated the cross-listing.

²⁶ Thomson SDC reports only 32 deals recognized under pooling in Europe since 1978, among 25,088 transactions with deal sizes greater than USD 1 million completed by public acquirers.

²⁷ We used a restricted set of control variables for this robustness check, due to the limited data availability for European firms.

²⁸ Some European countries allowed immediate acquisition goodwill write-offs in case of purchase accounting during the pooling period. The combination of purchase accounting with such immediate write-offs effectively lead purchase accounting to be more similar to pooling. Such an option could, therefore, compromise the validity of using European evidence as a counterfactual to the United States. Immediate write-offs were forbidden after 1998 (see Amel-Zadeh et al. 2015). Thus, we replicate panel F analysis starting after 1998. Our results are robust to this issue (available on request). We thank an anonymous referee for mentioning this issue.

²⁹ See “SEC Amends Issuer Common Stock Repurchase Safe Harbor,” December 19, 2003, <http://www.stblaw.com/docs/default-source/cold-fusion-existing-content/publications/pub392.pdf?sfvrsn=2>.

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